

CHAPTER 9

Counters

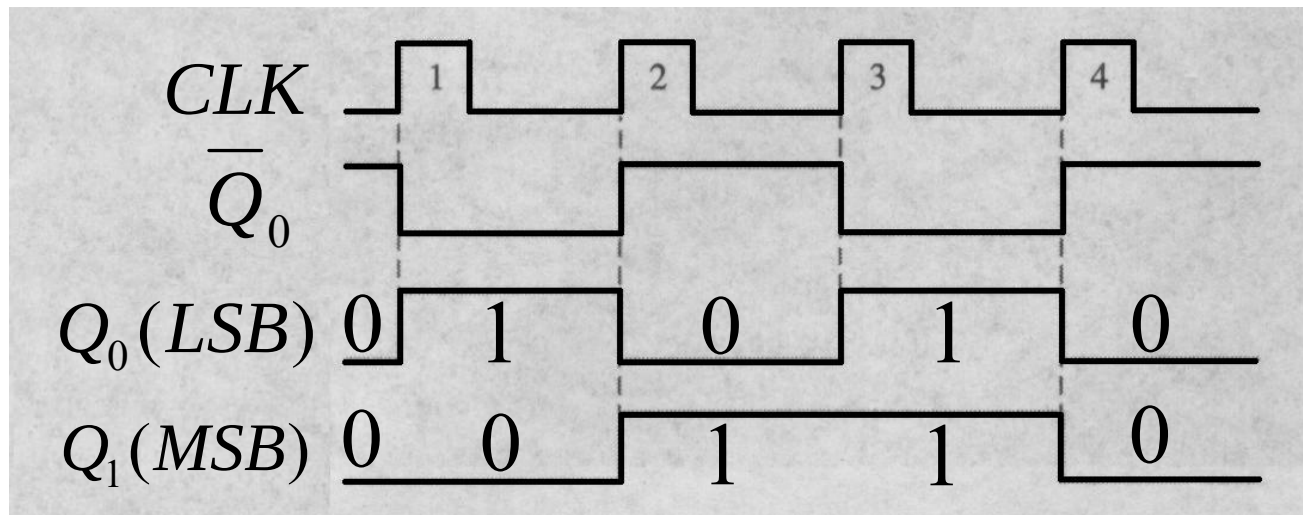
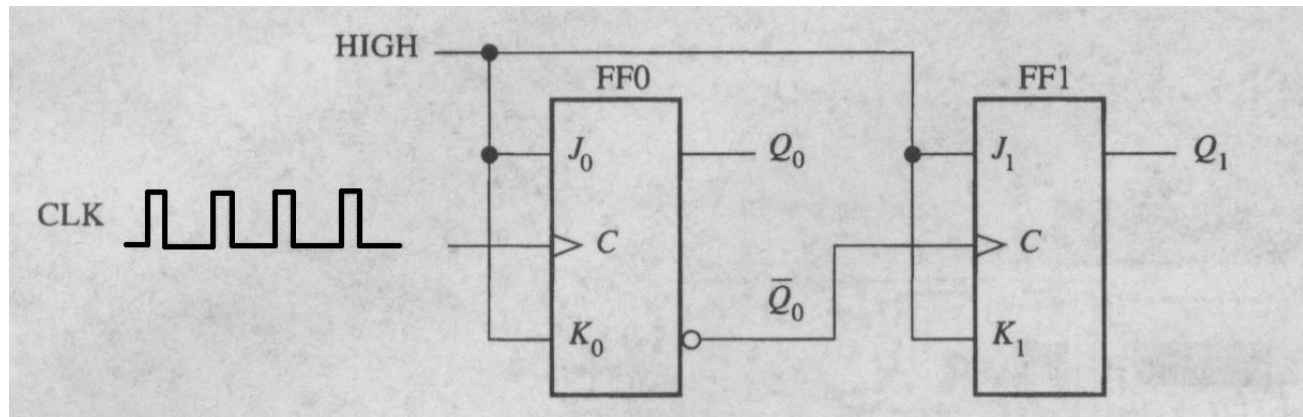
Synchronous and Asynchronous Counters

同步和异步计数器

- Flip-flops can be connected together to perform counting operations. Such a group of flip-flops is a counter.
- Counters are classified into two broad categories according to the way they are clocked: asynchronous and synchronous.
- The term asynchronous refers to events that do not have a fixed time relationship with each other and, generally, do not occur at the same time.
- In asynchronous counters, commonly called ripple counters, the first flip-flop is clocked by the external clock pulse and then each successive flip-flop is clocked by the output of the preceding flip-flop. So the flip-flops within an asynchronous counters do not change states at exactly the same time
- In synchronous counters, the clock input is connected to all of the flip-flops so that they are clocked simultaneously.

A 2-Bit Asynchronous Binary Counter

- This circuit consists of two flip-flops.
- The CLK is applied to the clock input of the first flip-flop. The second flip-flop is triggered by the output of the first flip-flop.

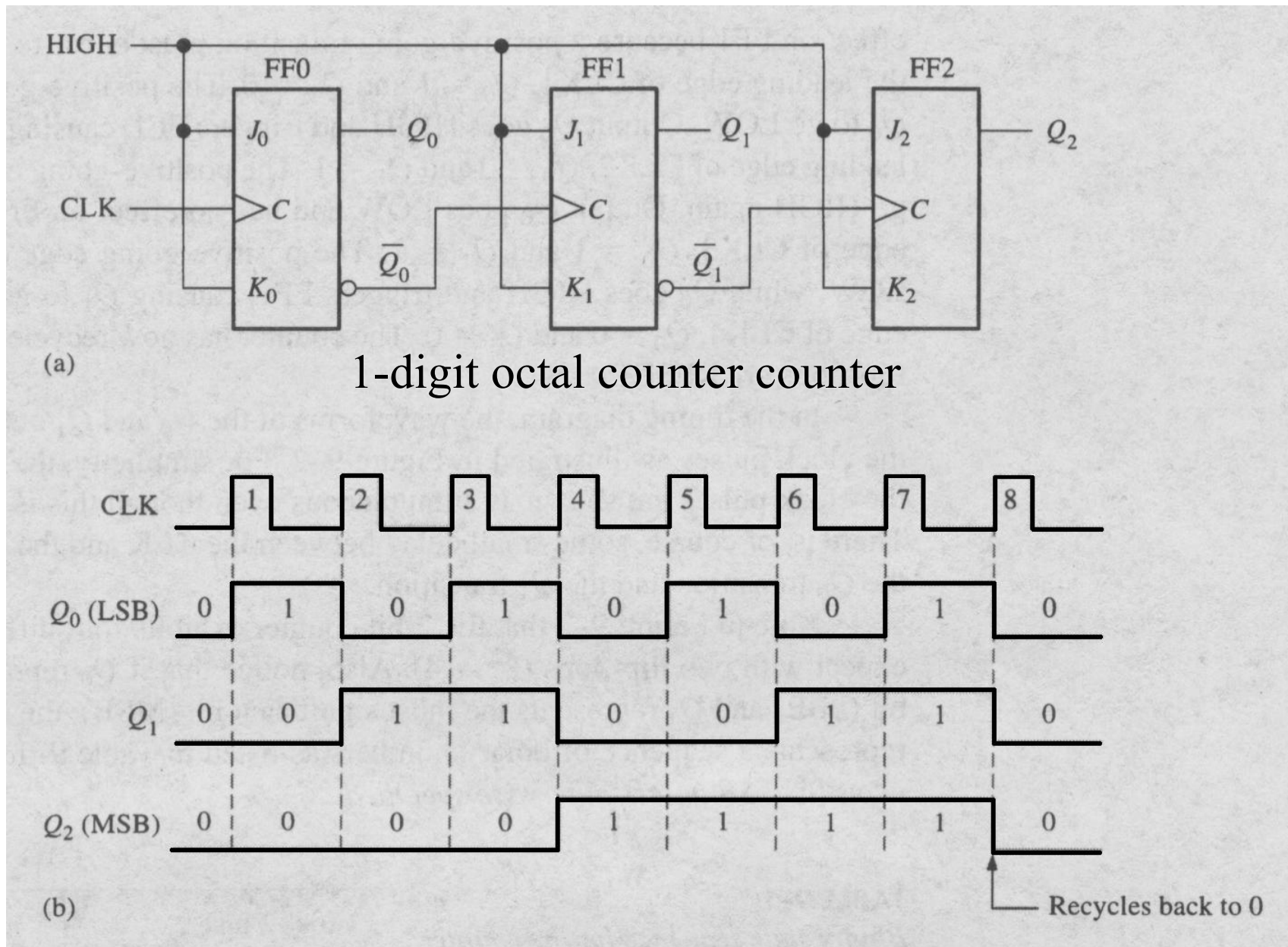


A 2-Bit Asynchronous Binary Counter

- This circuit is a 2-bit counter because it exhibits four different states (00, 01, 10, 11). In other words, it goes through a binary sequence (00, 01, 10, 11).
- It actually counts the number of clock pulses up to three, and on the fourth pulse it recycles to its original state.

Clock Pulse	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4 (recycles)	0	0

A 3-Bit Asynchronous Binary Counter

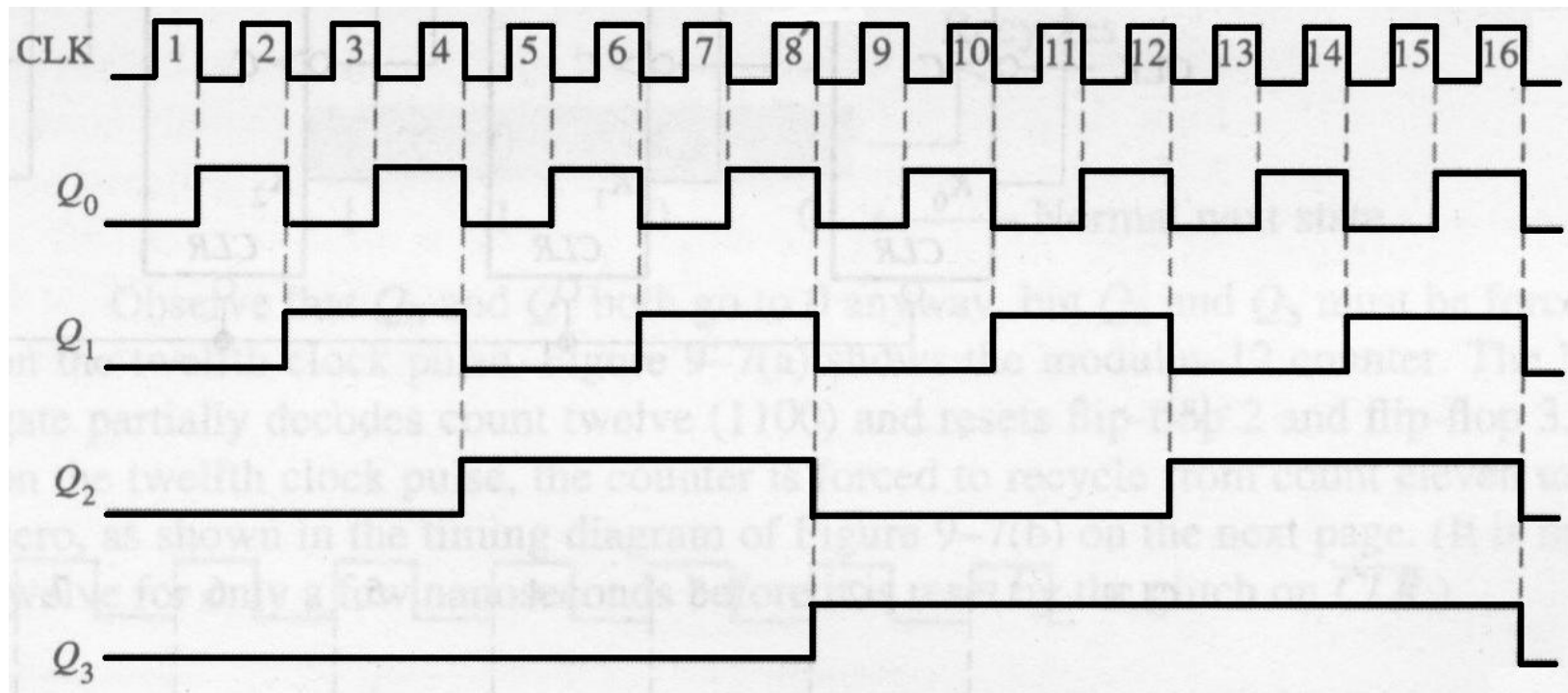
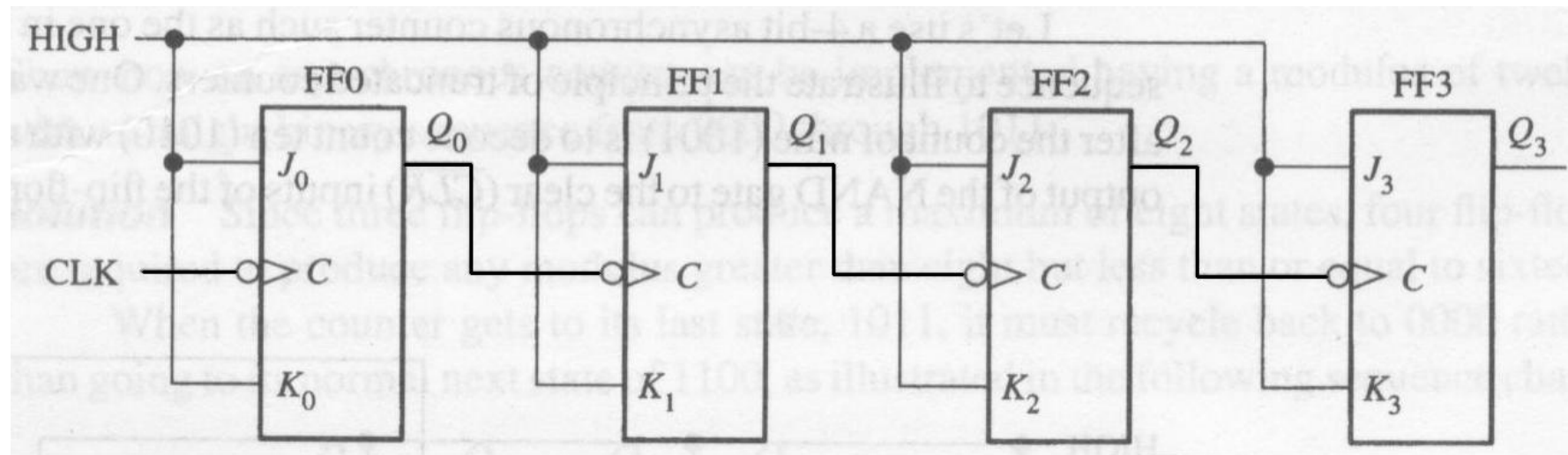


A 3-Bit Asynchronous Binary Counter

- A 3-bit counter goes through a binary sequence (000, 001, 010, 011, 100, 101, 110, 111).

Clock Pulse	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8 (recycles)	0	0	0

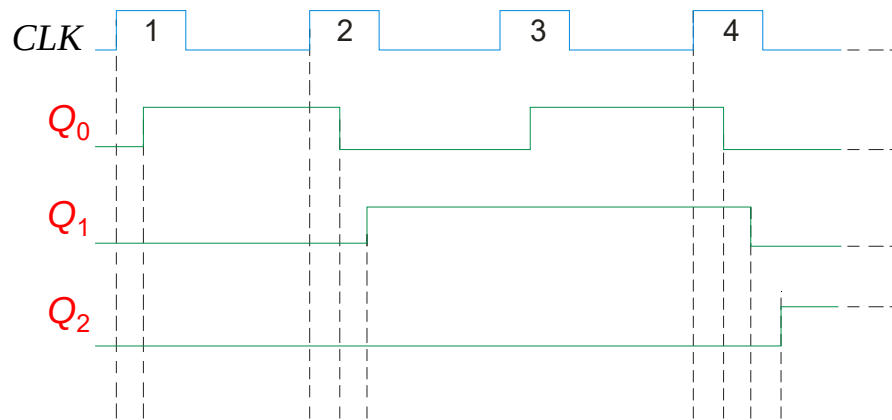
A 4-Bit Asynchronous Binary Counter



Asynchronous Binary Counter Propagation Delay

Asynchronous counters are sometimes called **ripple** counters, because the stages do not all change together. For certain applications requiring high clock rates, this is a major disadvantage.

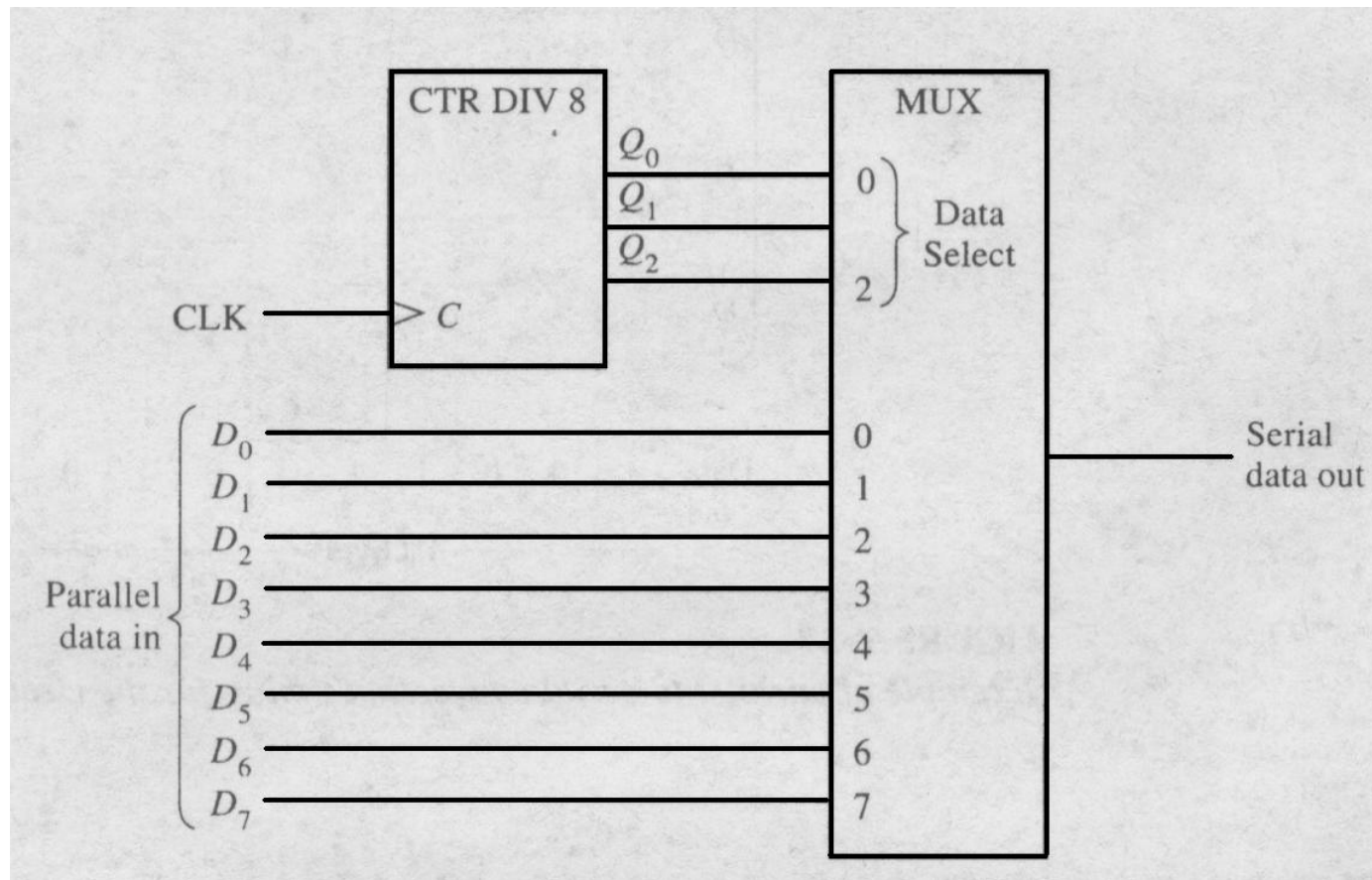
Notice how delays are cumulative as each stage in a counter is clocked later than the previous stage.



Q_0 is delayed by 1 propagation delay, Q_1 by 2 delays and Q_2 by 3 delays.

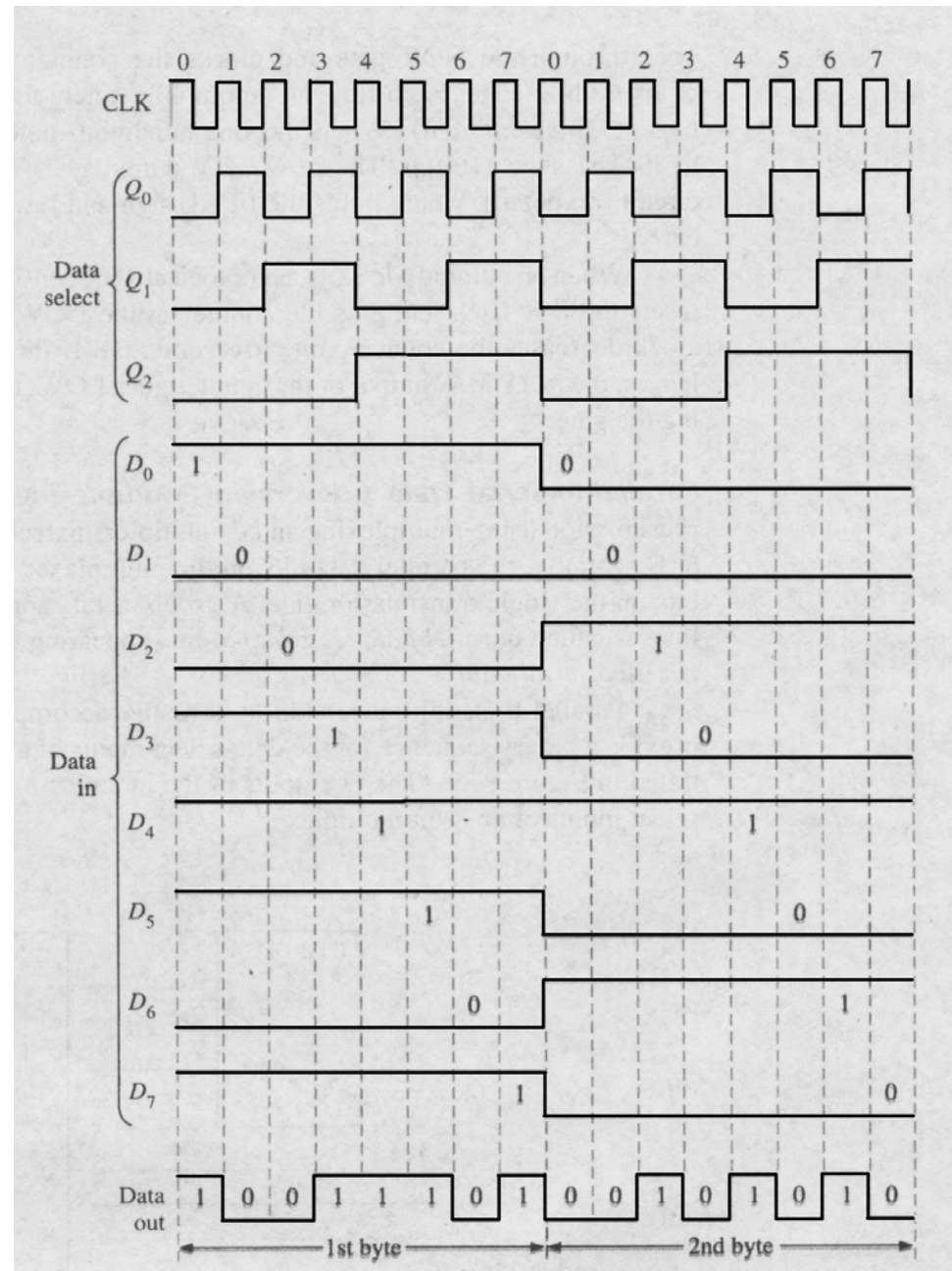
Parallel-to-Serial Data Conversion

- A group of bits appearing simultaneously on parallel lines is called parallel data. A group of bits appearing on a single line in a time sequence is called serial data.
- Parallel-to-serial conversion is normally accomplished by the use of a counter to provide a binary sequence for the data-select inputs of a mux.



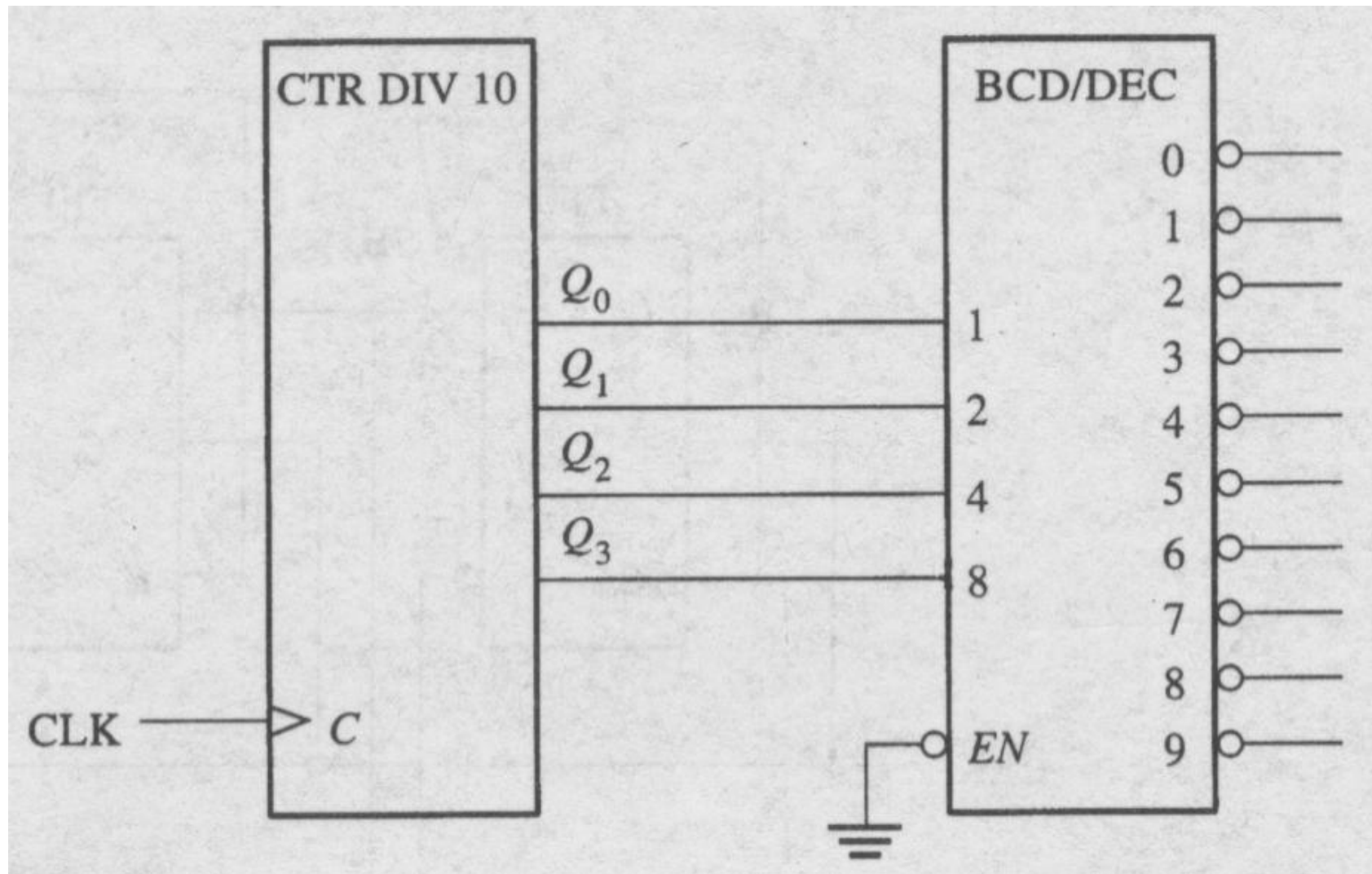
Parallel-to-Serial Data Conversion

- As the counter goes through a binary sequence from 0 to 7, each bit, beginning with D_0 , is sequentially selected and passed through the mux to the output line.



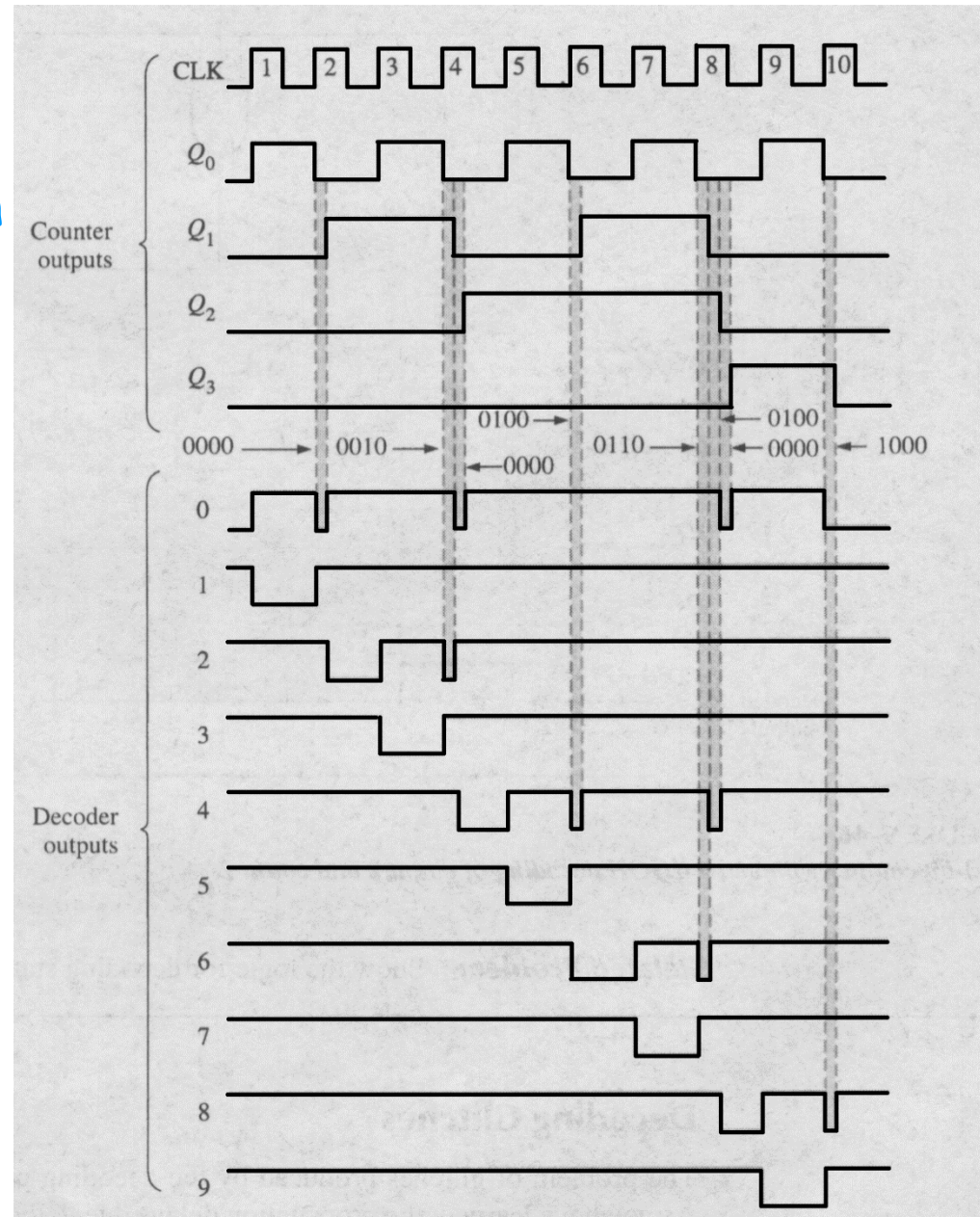
Decoding Glitches

- Some transitional state produce undesired voltage spikes of short duration (glitches) on the outputs of a decoder connected to the counter.



Decoding Glitches

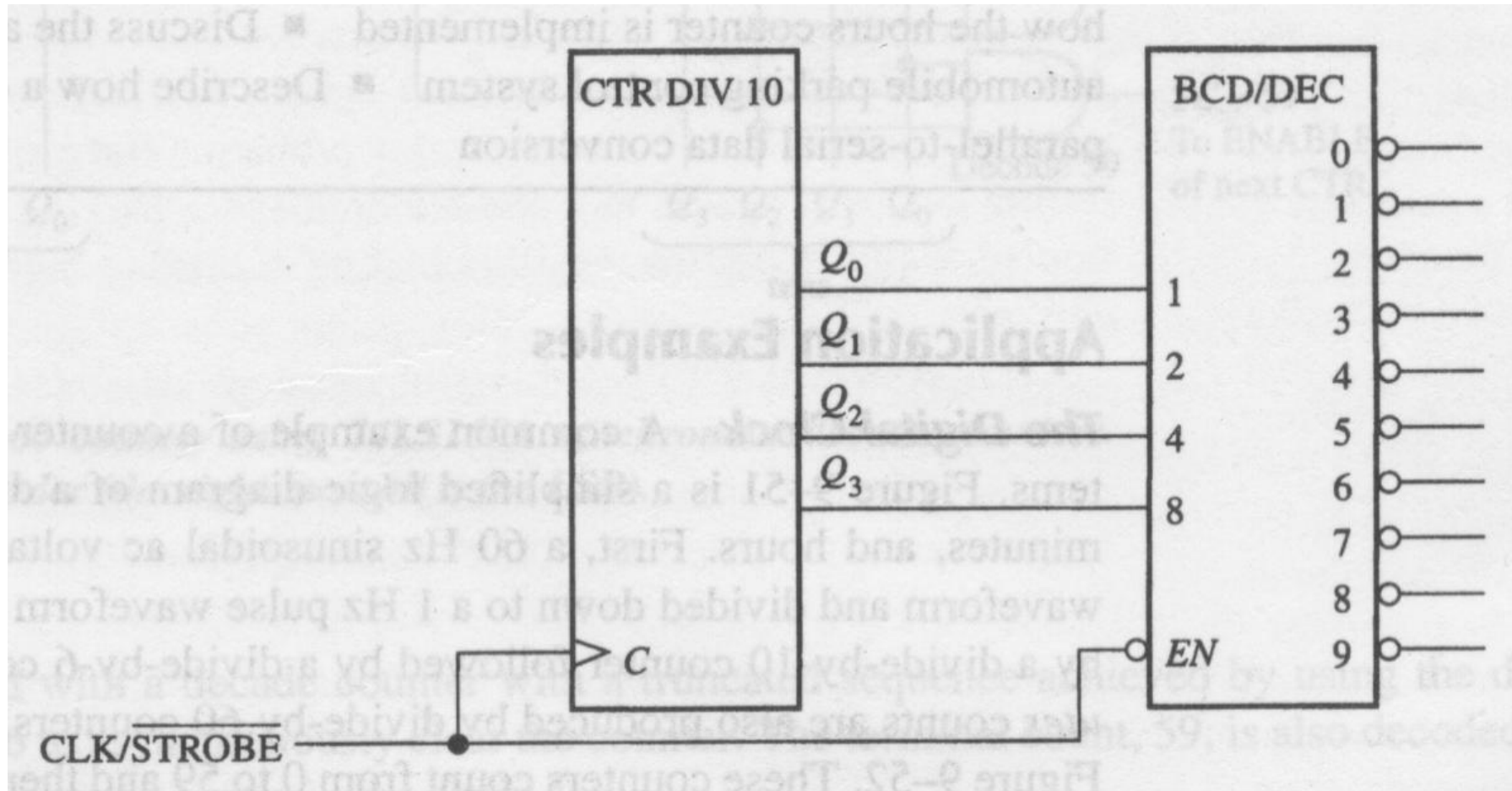
- Glitches result from propagation delays. The delays cause false states of short duration.



CLK	$Q_3 Q_2 Q_1 Q_0$
0	0000
1	0001
2	0000, 0010
3	0011
4	0010, 0000, 0100
5	0101
6	0100, 0110
7	0111
8	0110, 0100, 0000, 1000

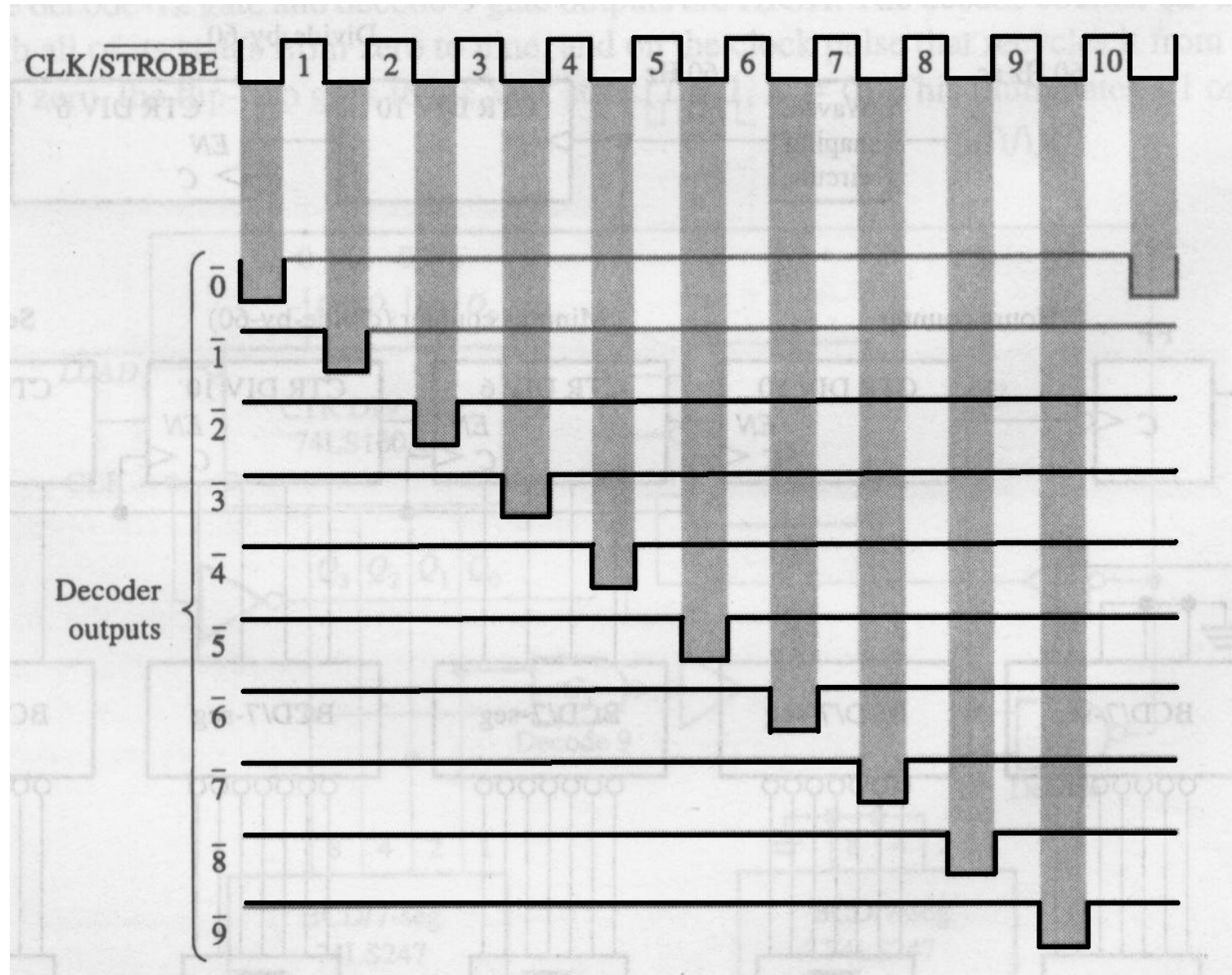
Using Strobe signal to eliminate Decoding Glitches

- One way to eliminate the glitches is to enable the decoded outputs at a time after the glitches have had time to disappear.



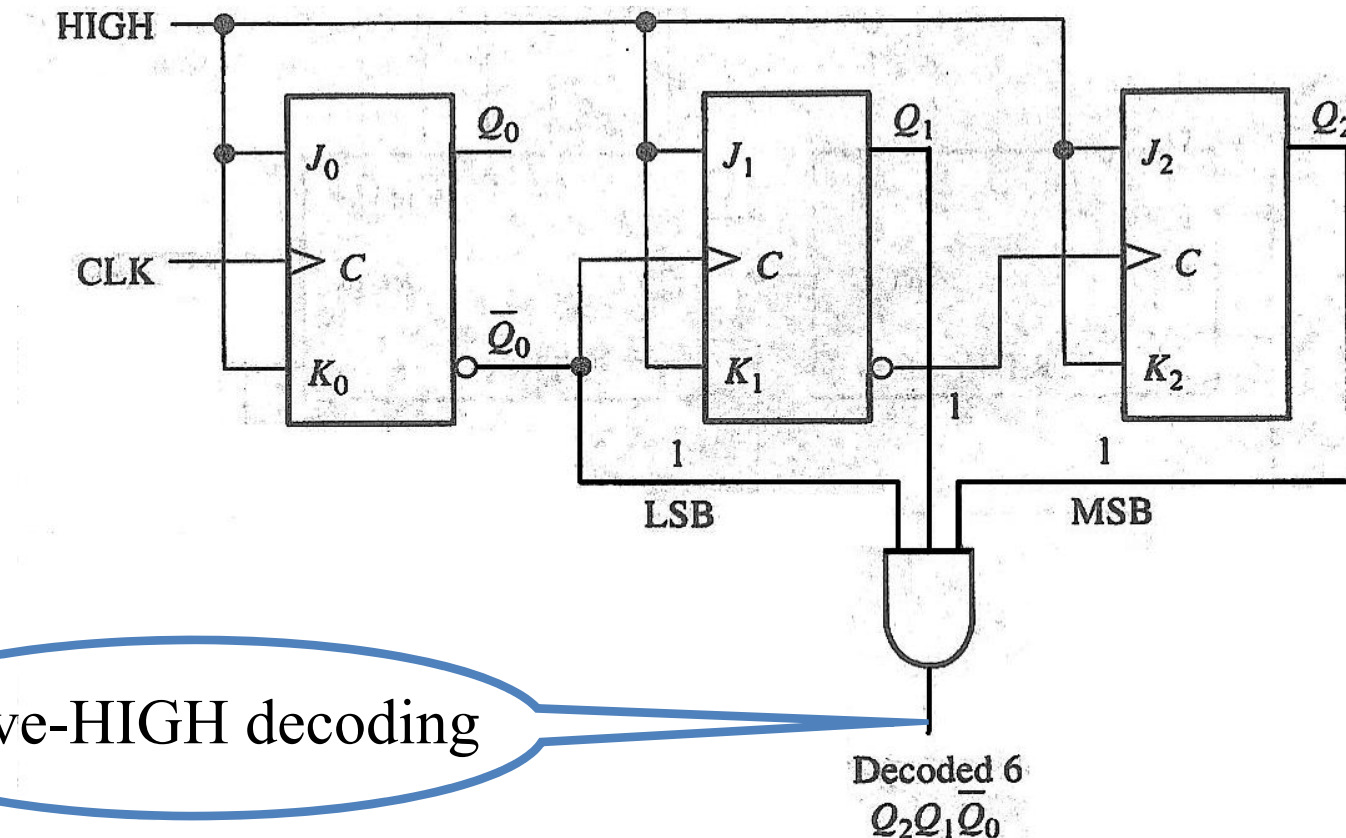
Using Strobe signal to eliminate Decoding Glitches

- For a positive edge triggered counter, the decoder is enabled when the CLK is LOW.



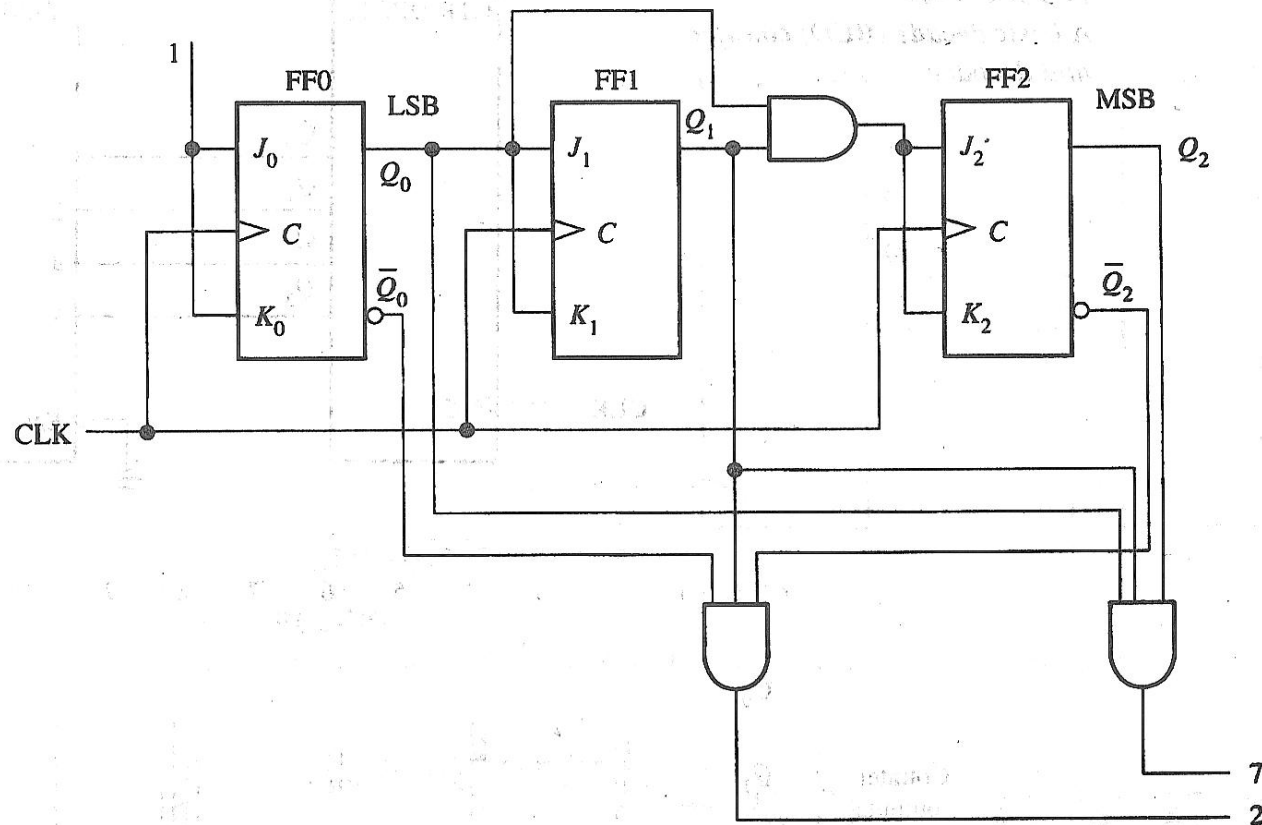
COUNTER DECODING

- In many applications, it is necessary that some or all of the counter states be decoded. The decoding of a counter involves using decoders to determine when the circuit is in a certain state in its sequence.
- Suppose we wish to decode state 6 (110) of a 3-bit counter.



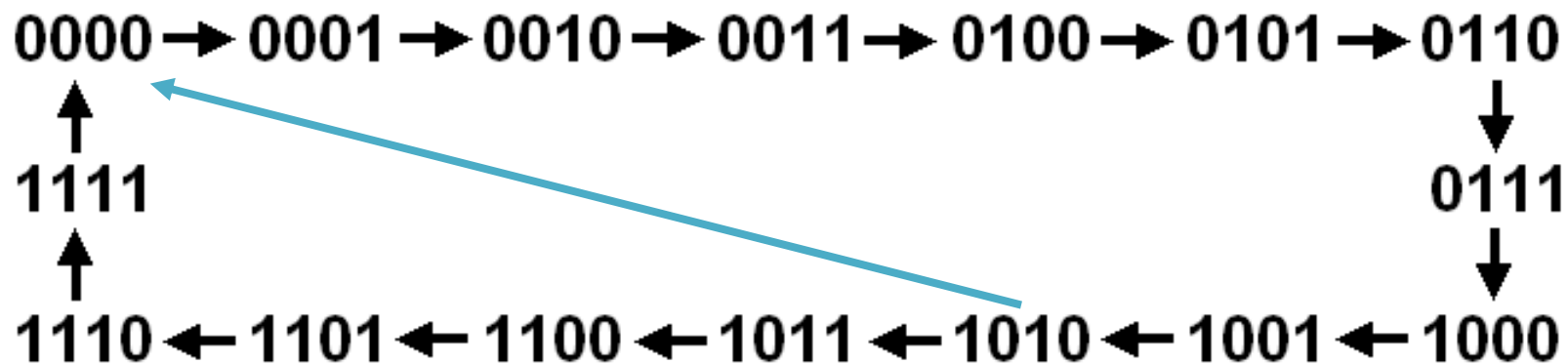
Decoding of Binary States 2 and 7

- A 3-bit counter with active-HIGH decoding of count 2 and count 7.

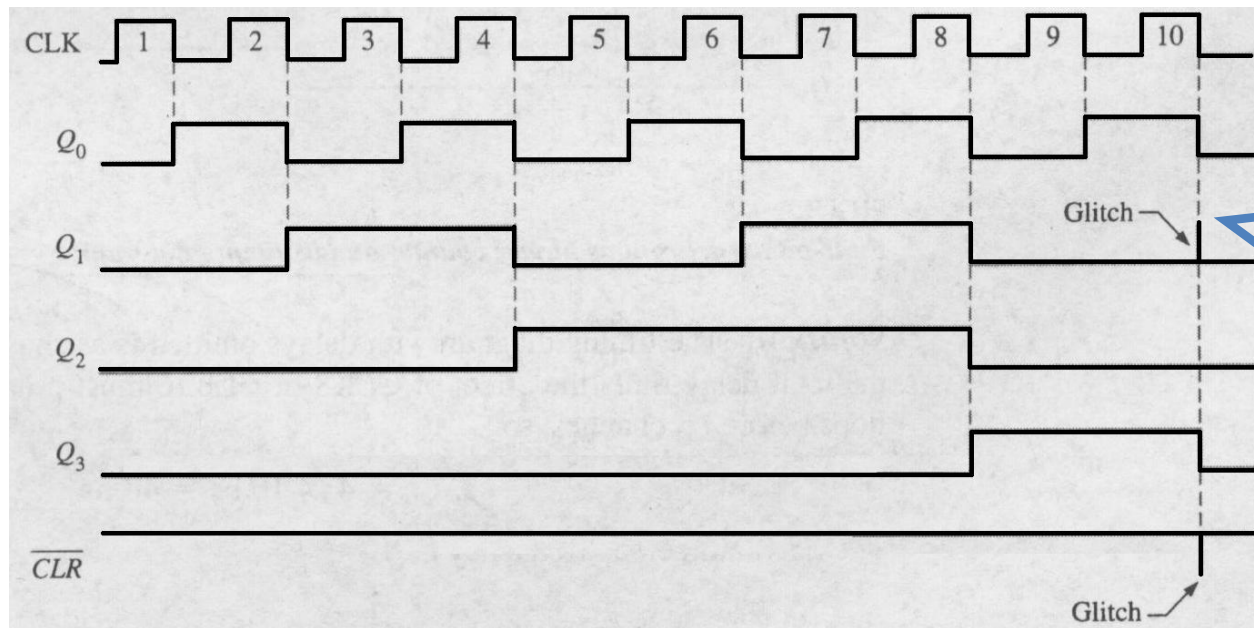
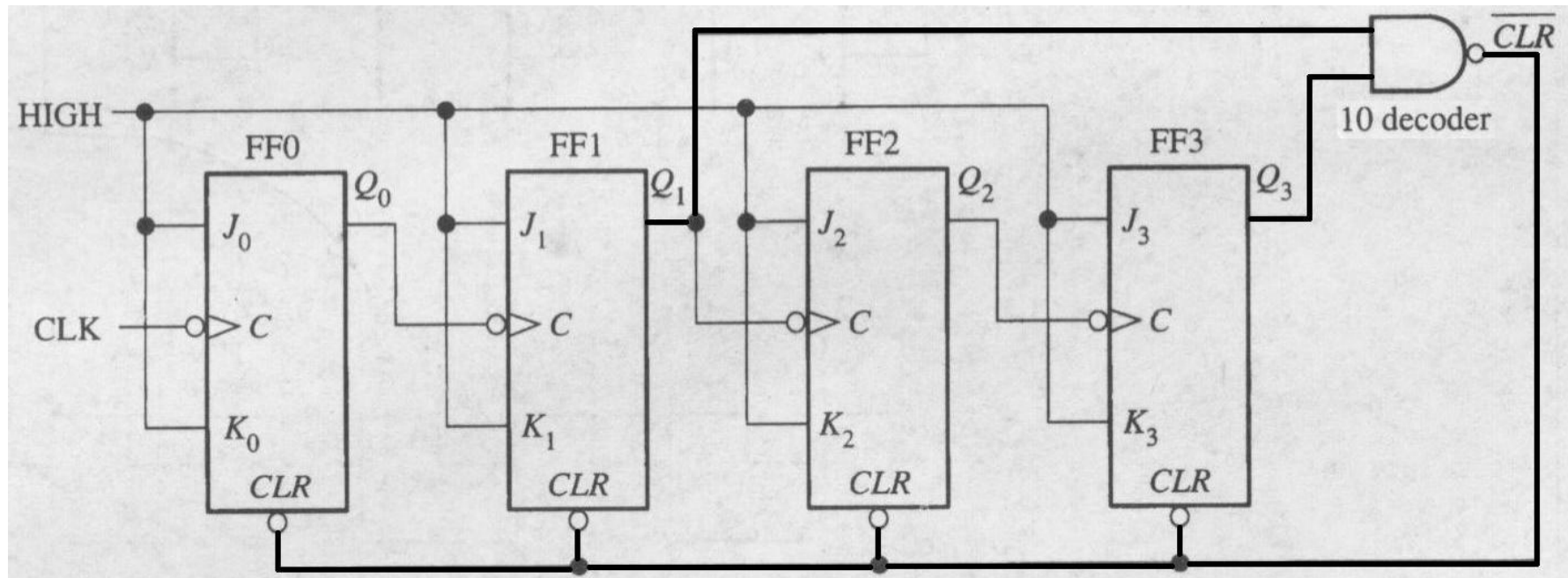


An Asynchronous Decade Counter

- The **modulus** of a counter is the number of unique states that the counter will sequence through. The maximum possible number of states (maximum modulus) of a counter is 2^n , where n is the number of flip-flops in the counter.
- Counters can also be designed to have a number of states in their sequence that is less than 2^n . The resulting sequence is called a truncated sequence.
- One common modulus for counters with truncated sequences is ten (called MOD 10). Counters with ten states in their sequence are called **decade counters**. A decade counter with a count sequence of 0 through 9 is a **BCD decade counter**.
- To obtain a truncated sequence, it is necessary to force the counter to recycle before going through all of its possible states.

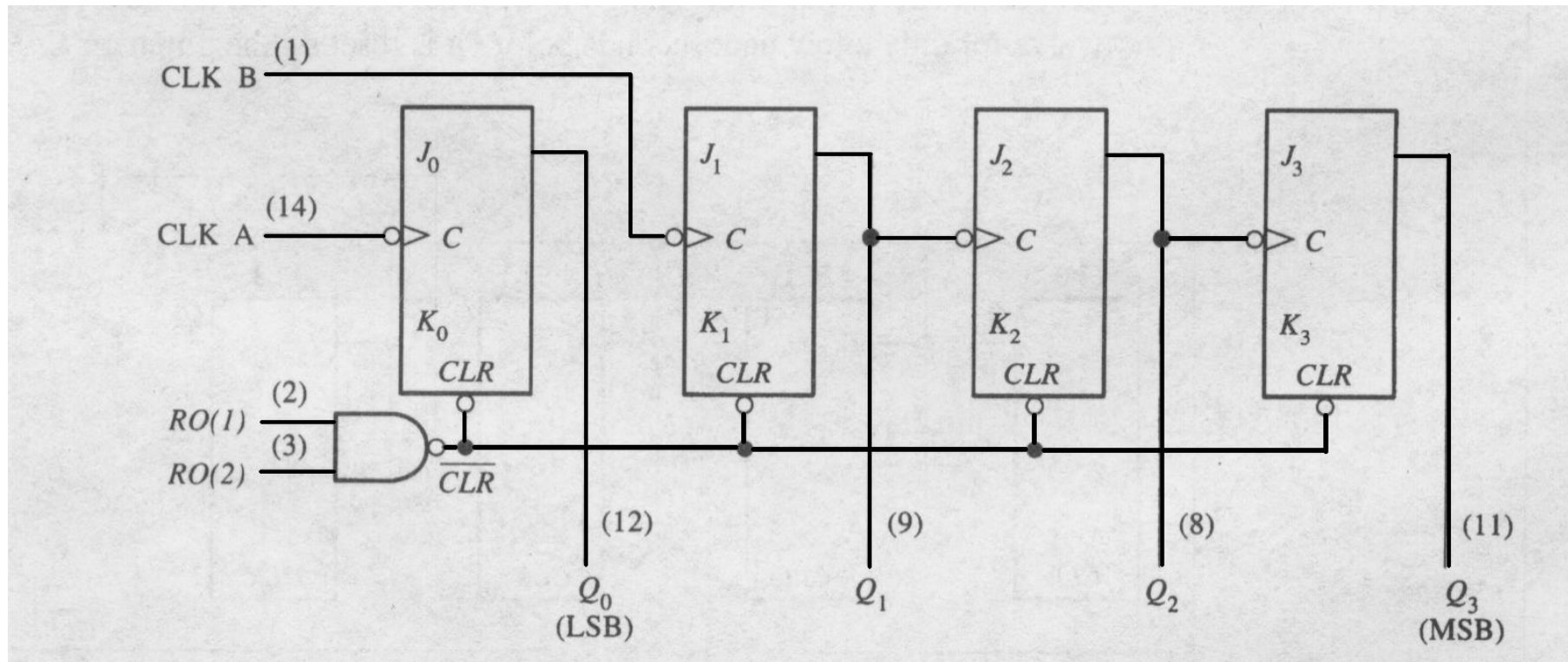


An Asynchronous Decade Counter



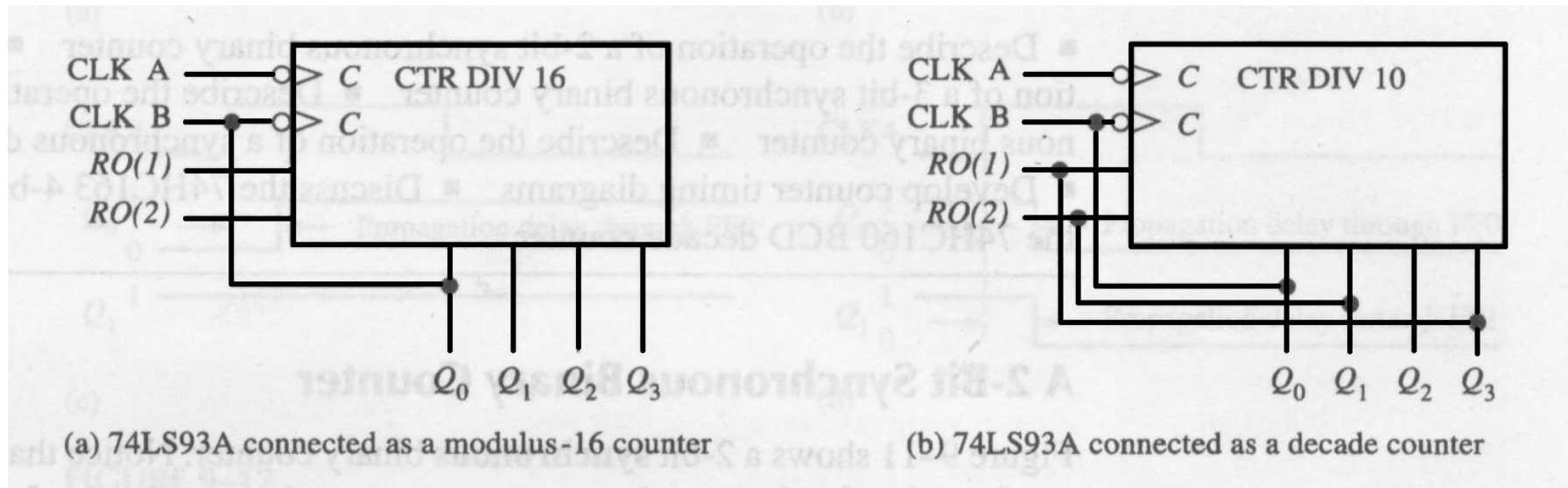
Glitch(毛刺、瑕疵):
a false or spurious
electronic signal.

74LS93A: 4-Bit Asynchronous Binary Counter



- It can be used as a divide-by-2 device if only the single FF is used.
- It can be used as a modulus-8 counter if only the 3-bit counter portion is used.

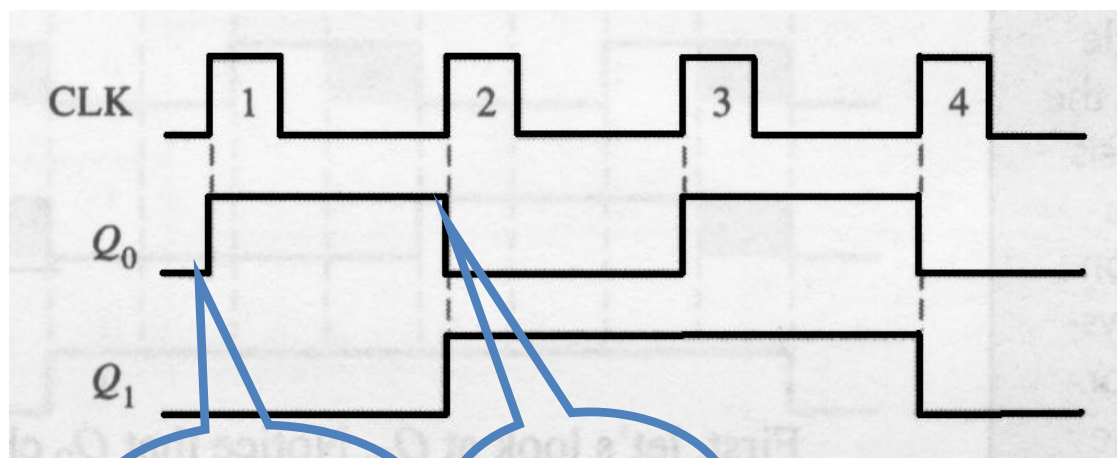
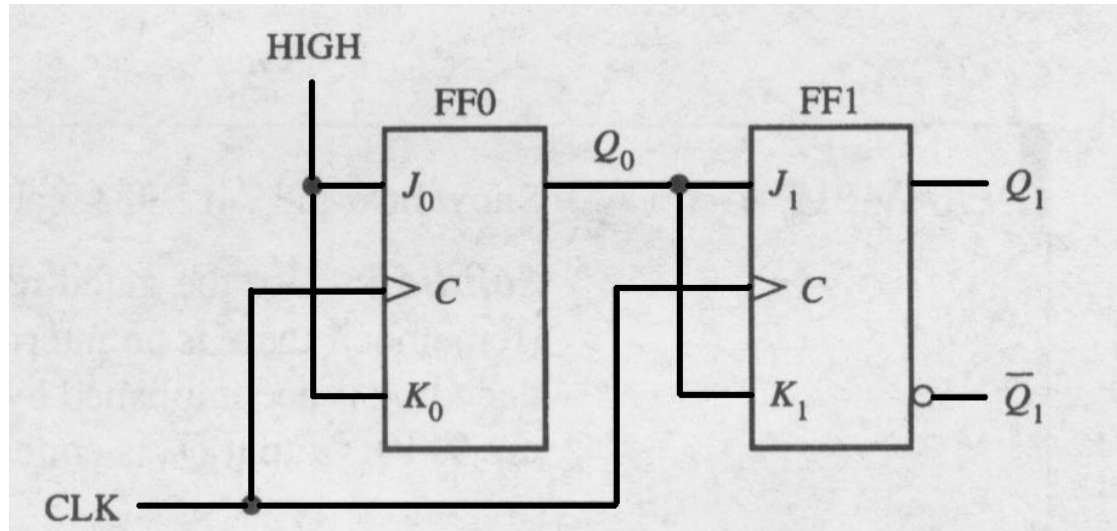
74LS93A: 4-Bit Asynchronous Binary Counter



- It can be used as a modulus-16 counter by connecting the Q₀ output to the CLK B input.
- It can be used as a decade counter by using the gated reset inputs for partial decoding of count ten (1010).

8-3 SYNCHRONOUS COUNTER OPERATION

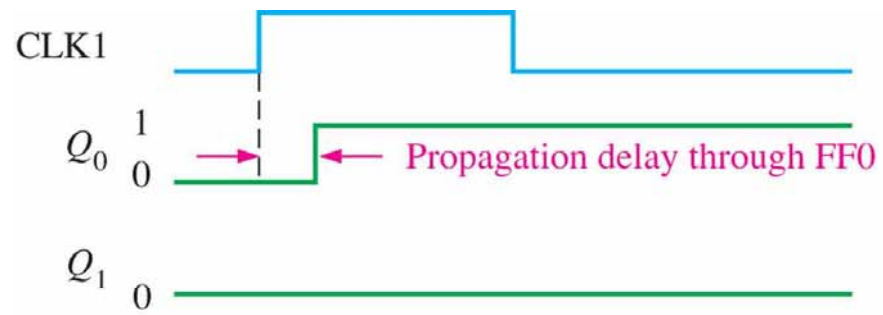
- Synchronous means that all the FFs in the counter are clocked at the same time by a common clock pulse.



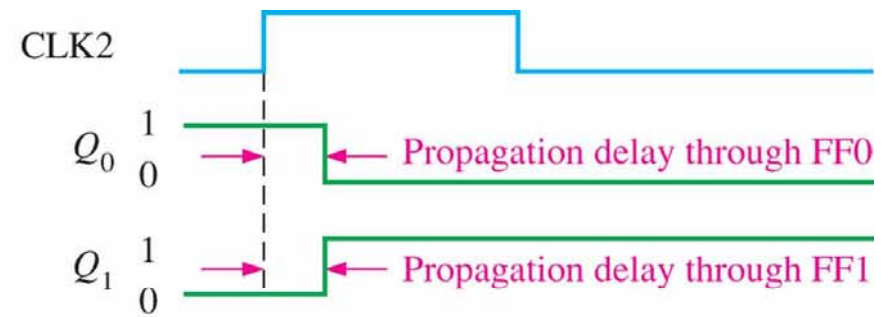
No-change condition

Toggle condition

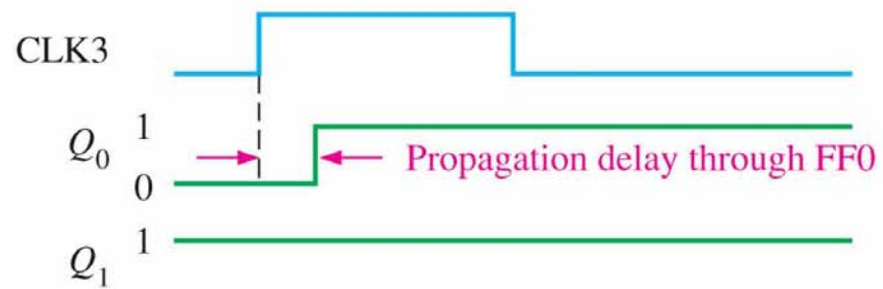
- FF0 is always connected for toggle operation.
- When the leading edge of CLK occurs, FF1 toggles if Q_0 is 1, otherwise it remains in its present state.
- There is a propagation delay from the triggering edge of the clock pulse until the Q output actually makes a transition.



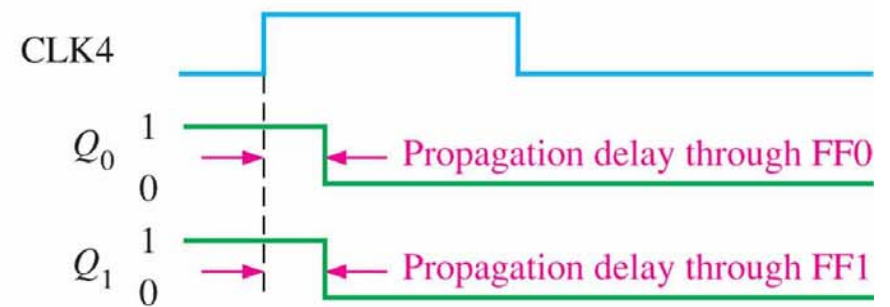
(a)



(b)

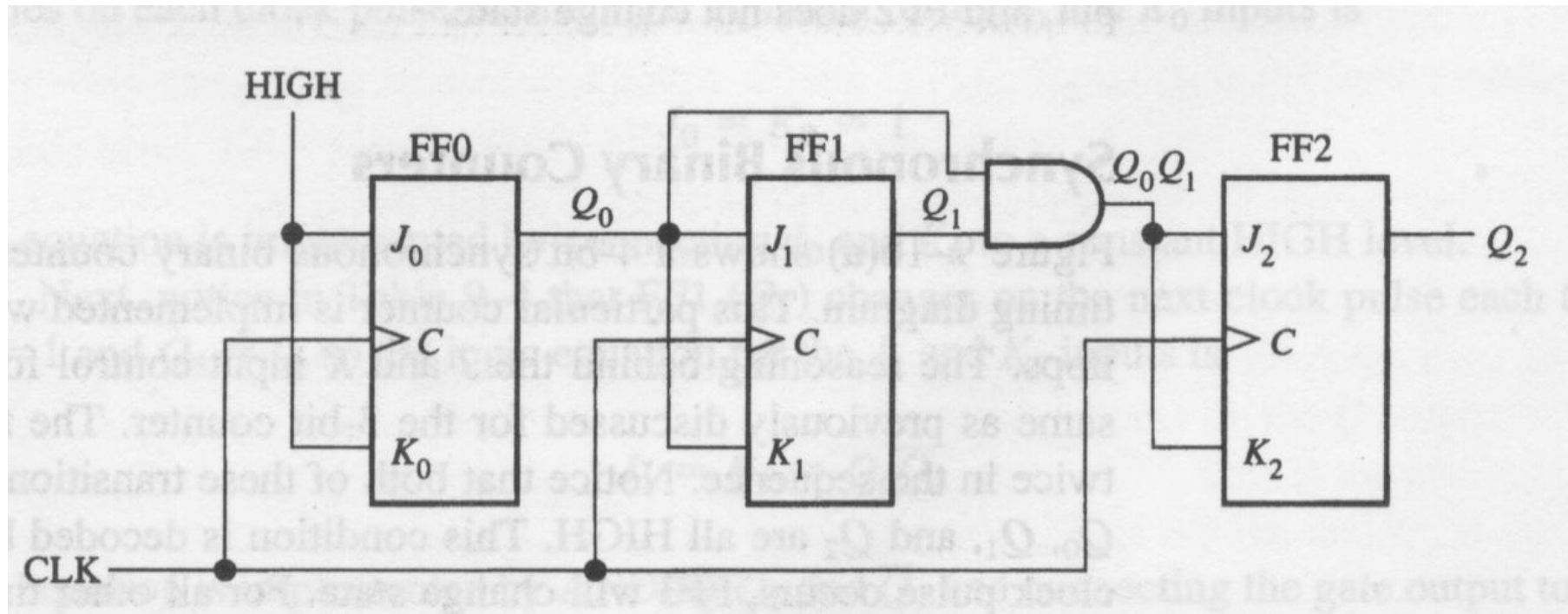


(c)



(d)

A 3-Bit Synchronous Binary Counter



$$J_0 = K_0 = 1$$

- FF0 is held in the toggle mode by constant HIGHs on its data inputs.

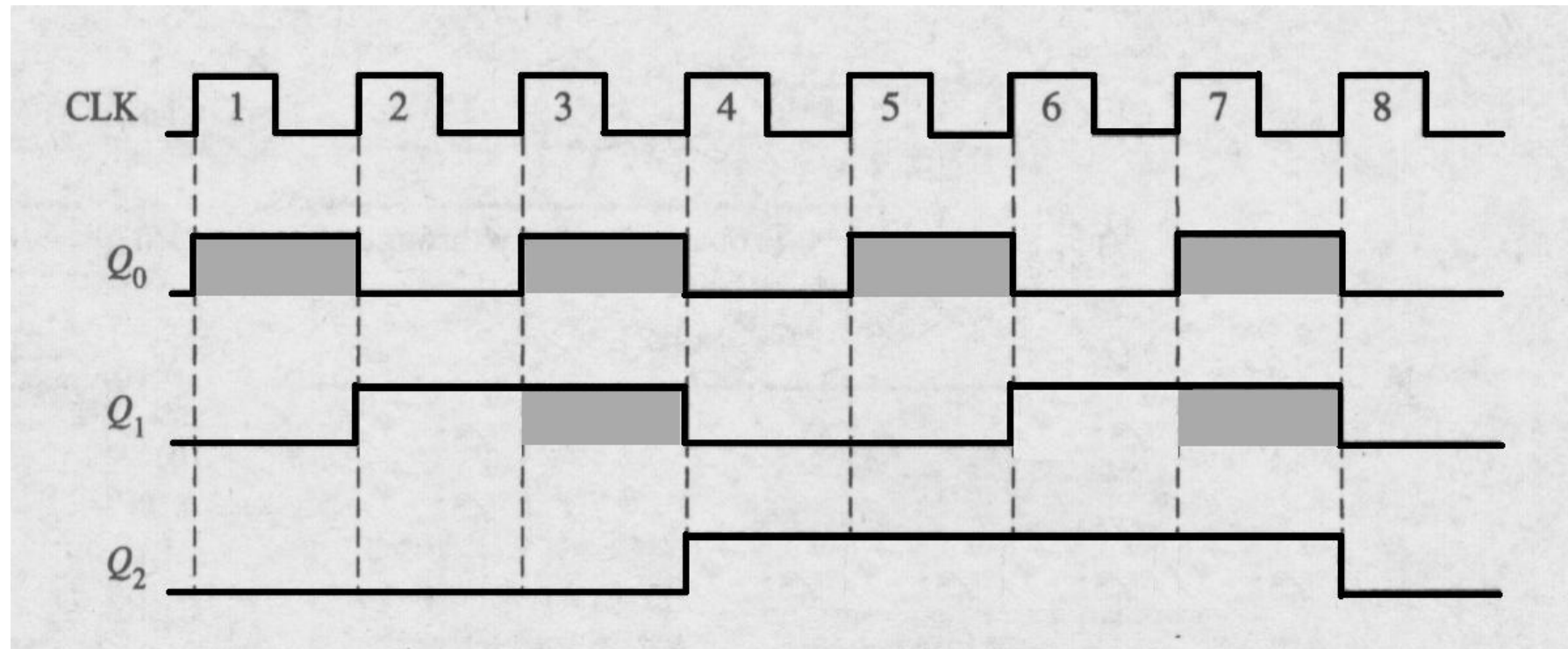
$$J_1 = K_1 = Q_0$$

- When Q₀ is a 1 FF1 is in the toggle mode.

$$J_2 = K_2 = Q_0 Q_1$$

- When Both Q₀ and Q₁ are HIGH, FF2 is in the toggle mode.

A 3-Bit Synchronous Binary Counter



$$J_0 = K_0 = 1$$

- FF0 is held in the toggle mode by constant HIGHs on its data inputs.

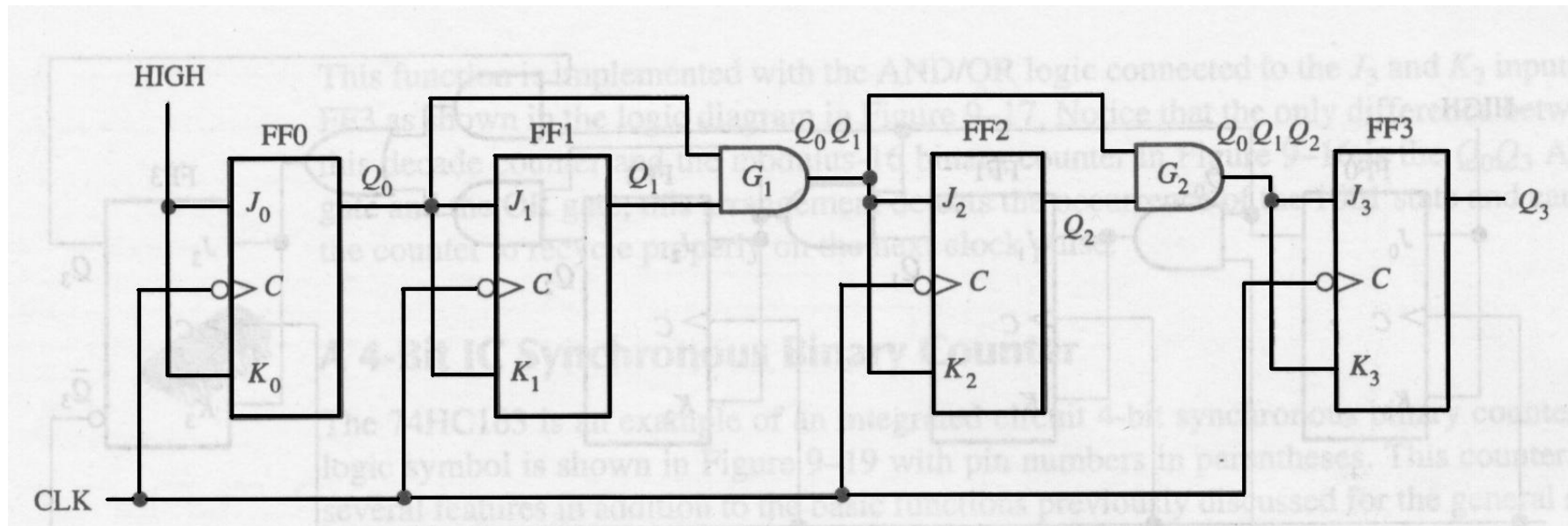
$$J_1 = K_1 = Q_0$$

- When Q_0 is a 1 FF1 is in the toggle mode.

$$J_2 = K_2 = Q_0 Q_1$$

- When Both Q_0 and Q_1 are HIGH, FF2 is in the toggle mode.

A 4-Bit Synchronous Binary Counter



$$J_0 = K_0 = 1$$

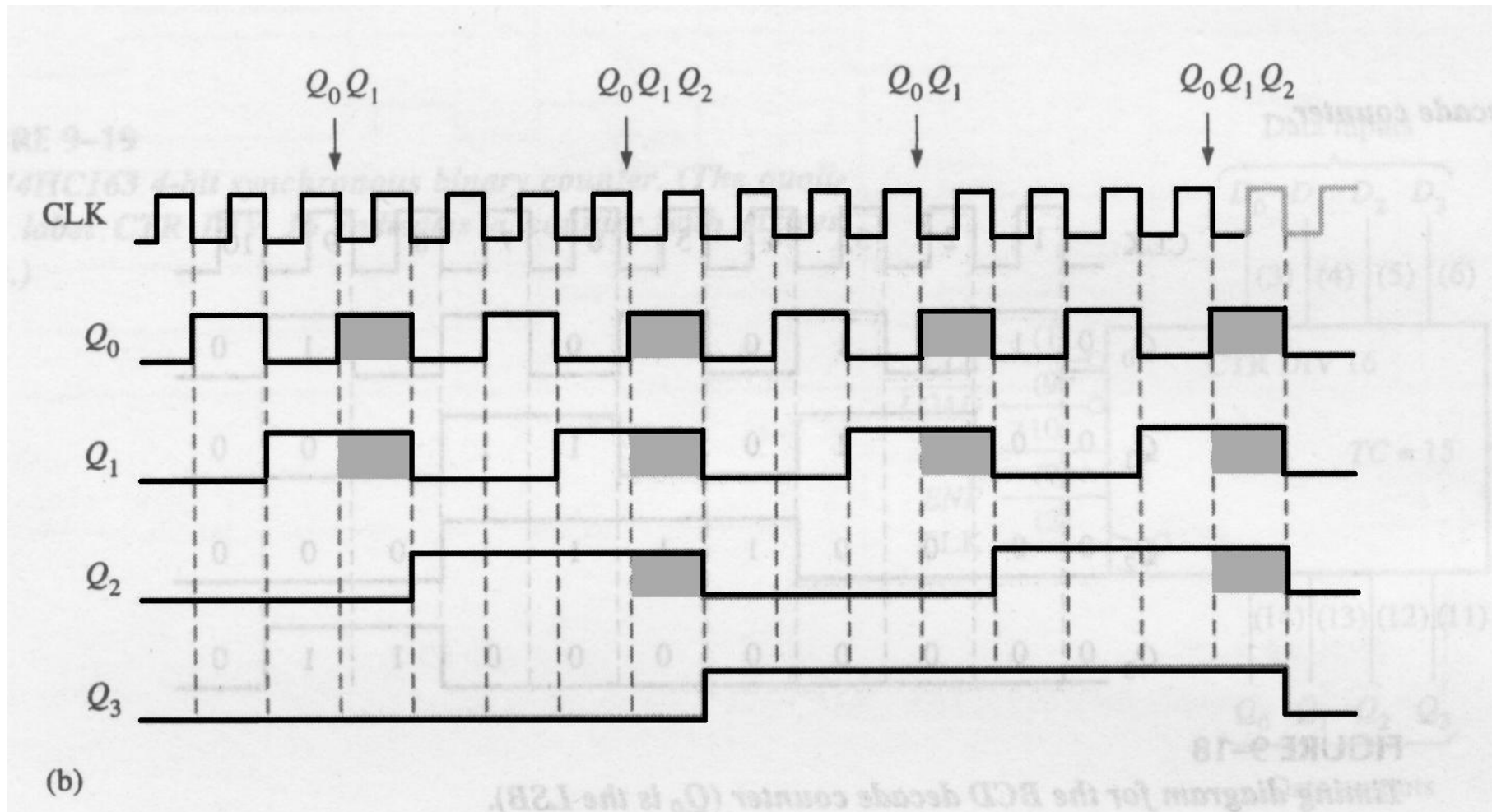
$$J_1 = K_1 = Q_0$$

$$J_2 = K_2 = Q_0 Q_1$$

$$J_3 = K_3 = Q_0 Q_1 Q_2$$

- FF0 is held in the toggle mode by constant HIGHs on its data inputs.
- When Q_0 is a 1 FF1 is in the toggle mode.
- When Both Q_0 and Q_1 are HIGH, FF2 is in the toggle mode.
- When Q_0 , Q_1 and Q_2 are all HIGH, FF3 is in the toggle mode.

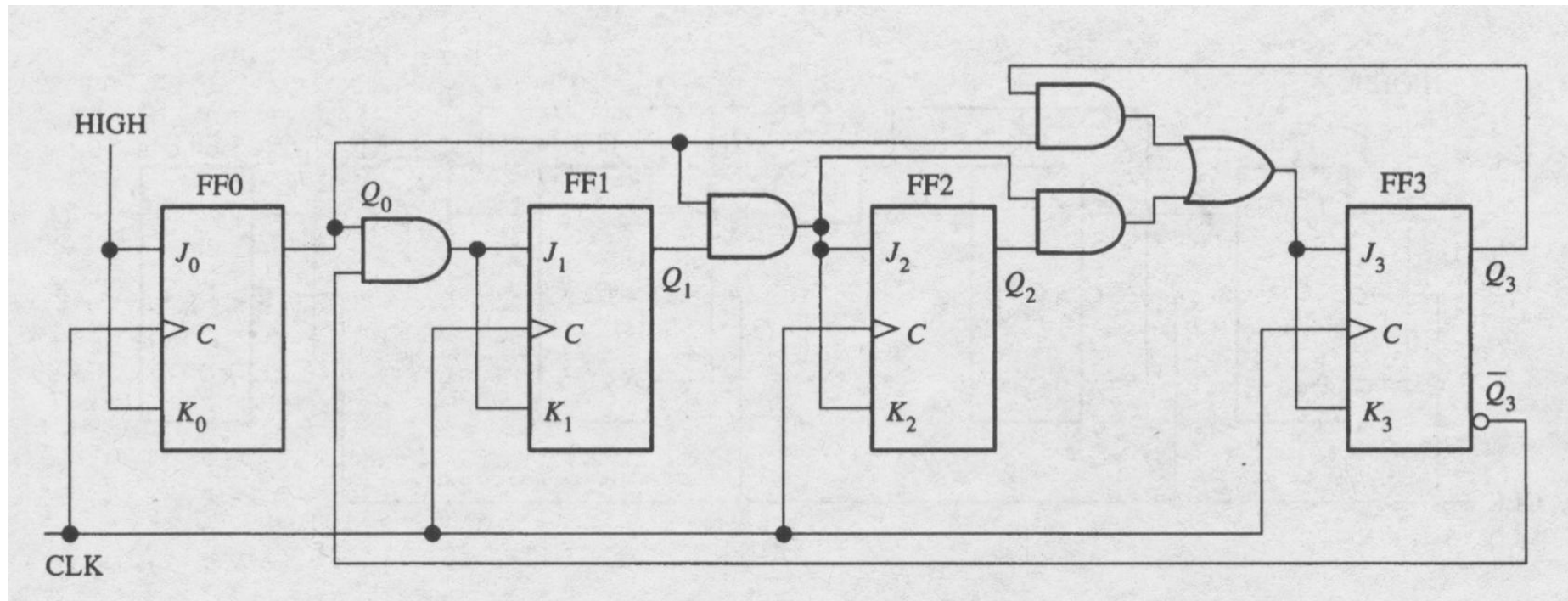
A 4-Bit Synchronous Binary Counter



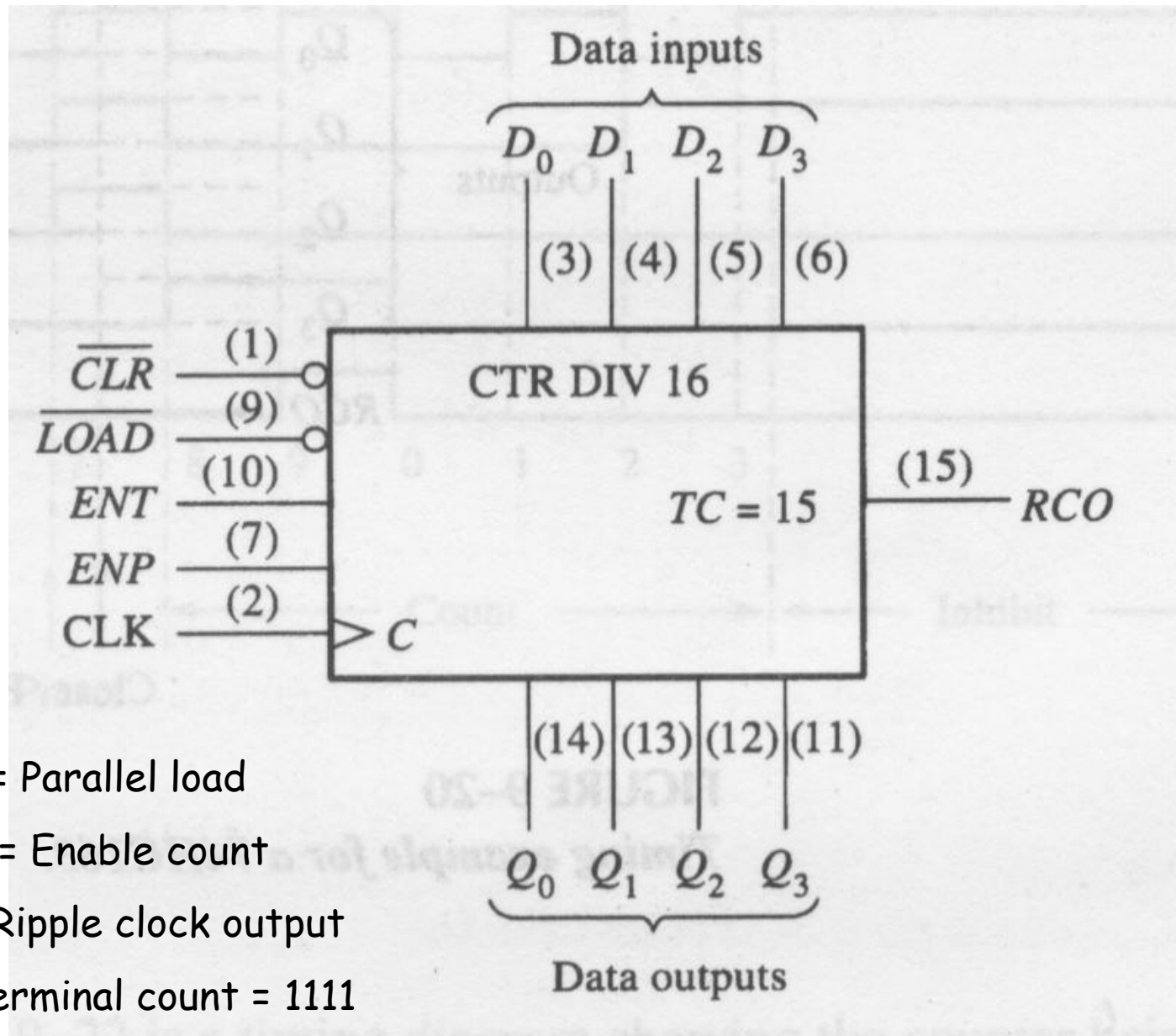
A Synchronous Decade Counter

Clock Pulse	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10 (recycles)	0	0	0	0

A Synchronous Decade Counter



74HC163: 4-Bit Synchronous Binary Counter



LOAD = Parallel load

EP, ET = Enable count

RCO = Ripple clock output

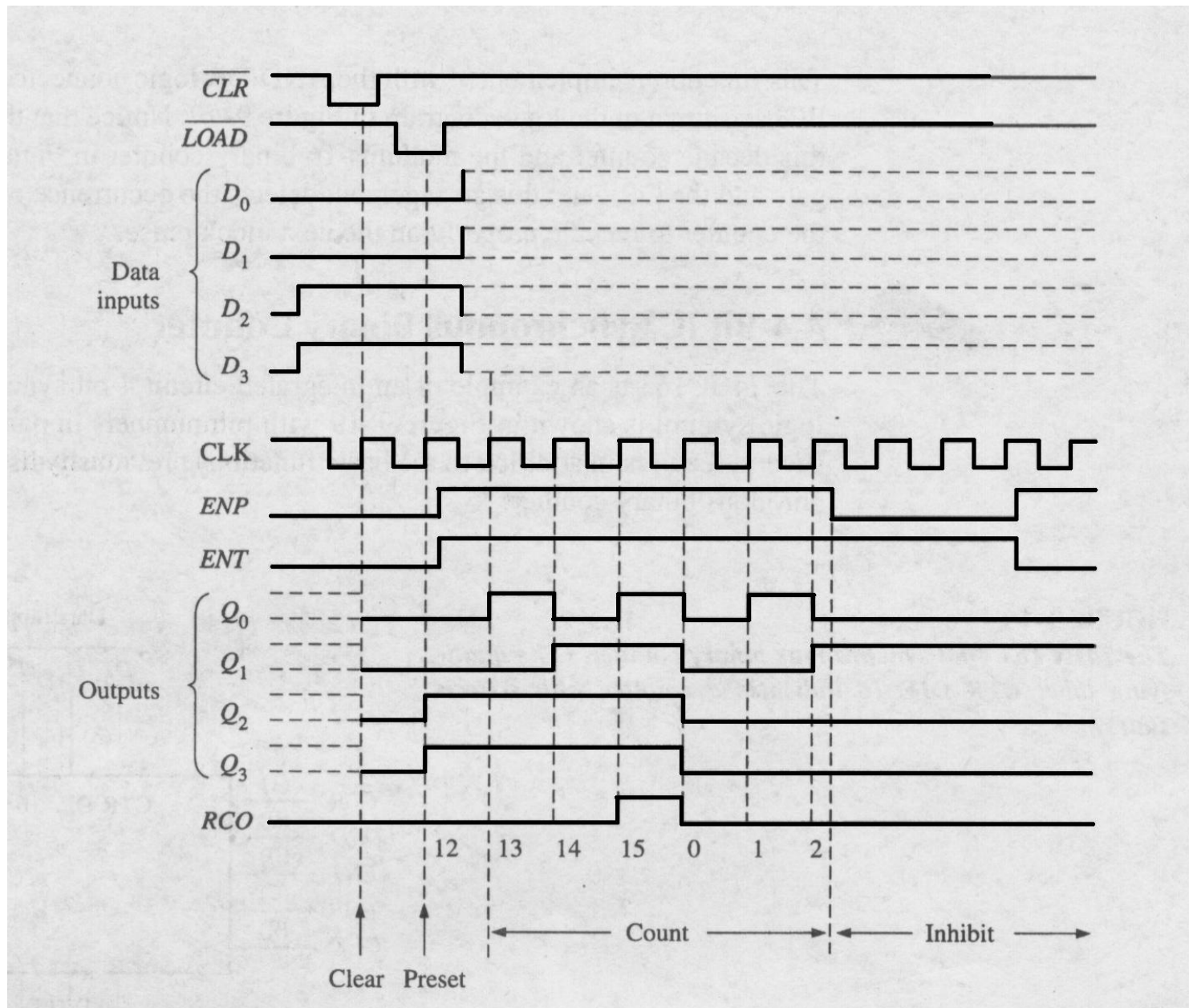
TC = Terminal count = 1111

Function Table for 74HC163

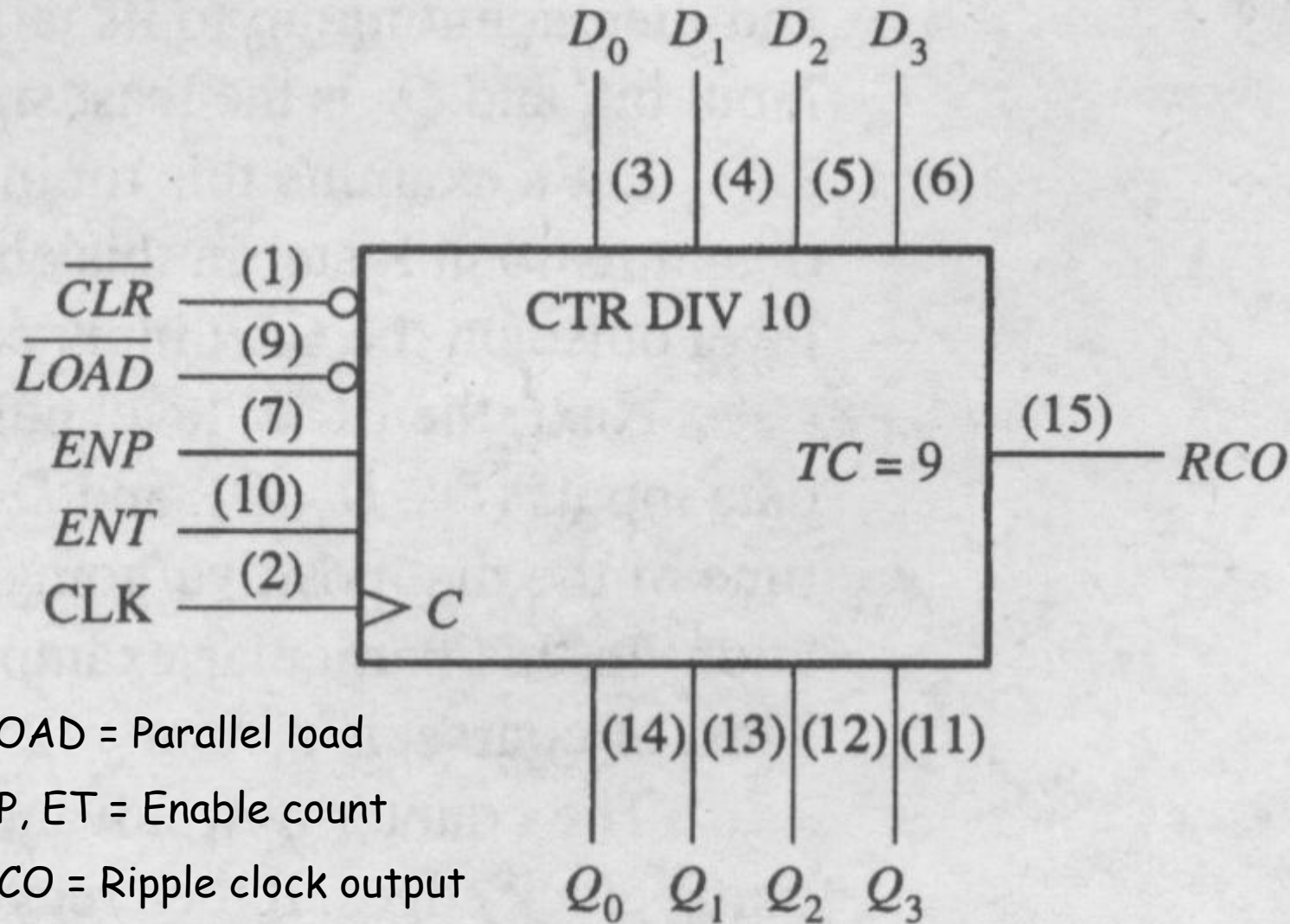
Operating Mode	$\overline{\text{CLR}}$	CLK	ENP	ENT	$\overline{\text{LOAD}}$	D	Q	RCO
Reset (Clear)	0	↑	X	X	X	X	0	0
Parallel load	1	↑	X	X	0	D	D	0
Count	1	↑	1	1	1	X	Count	†
Inhibit	1	X	0	X	1	X	Q	†
	1	X	X	0	1	X	Q	0

† The RCO output is 1 when ENT is 1 and the counter is at TC (1111).

Timing Example for a 74HC163



74HC160: Synchronous BCD Decade Counter



LOAD = Parallel load

EP, ET = Enable count

RCO = Ripple clock output

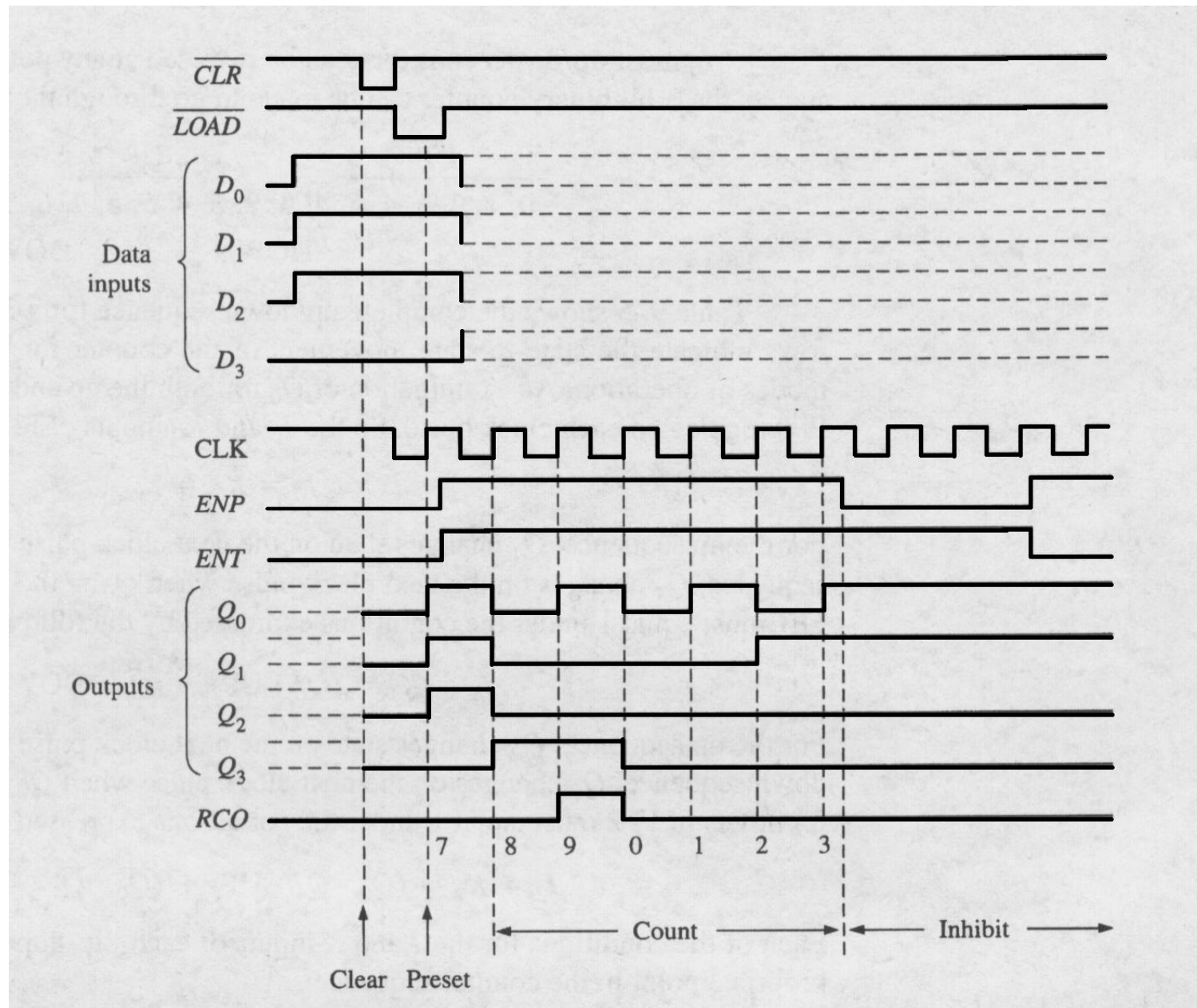
TC = Terminal count = 1001

Function Table for 74HC160

Operating Mode	$\overline{\text{CLR}}$	CLK	ENP	ENT	$\overline{\text{LOAD}}$	D	Q	RCO
Reset (Clear)	0	X	X	X	X	X	0	0
Parallel load	1	↑	X	X	0	D	D	0
Count	1	↑	1	1	1	X	Count	†
Inhibit	1	X	0	X	1	X	Q	†
	1	X	X	0	1	X	Q	0

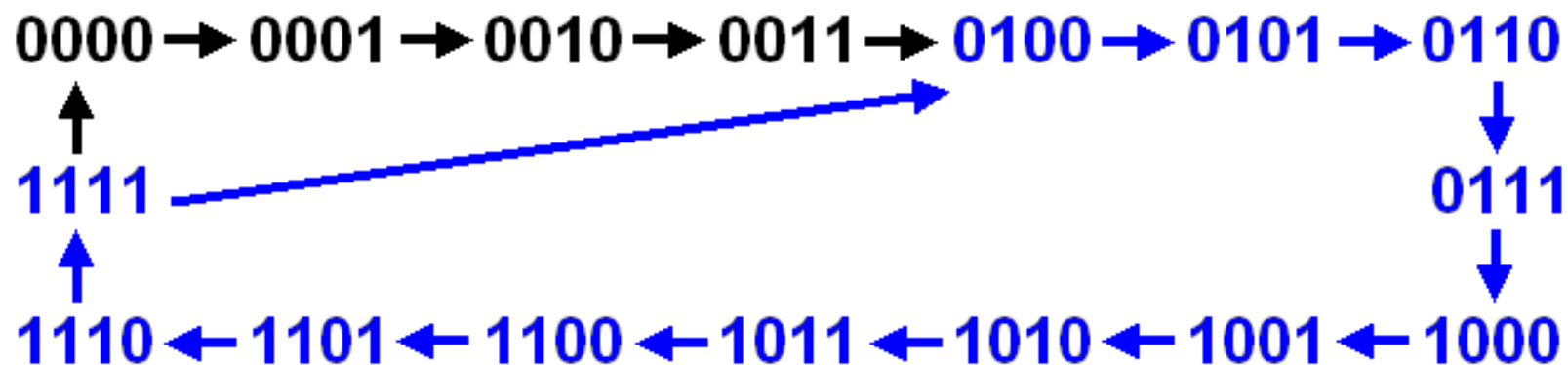
† The RCO output is 1 when ENT is 1 and the counter is at TC (1001)

Timing Example for a 74HC160

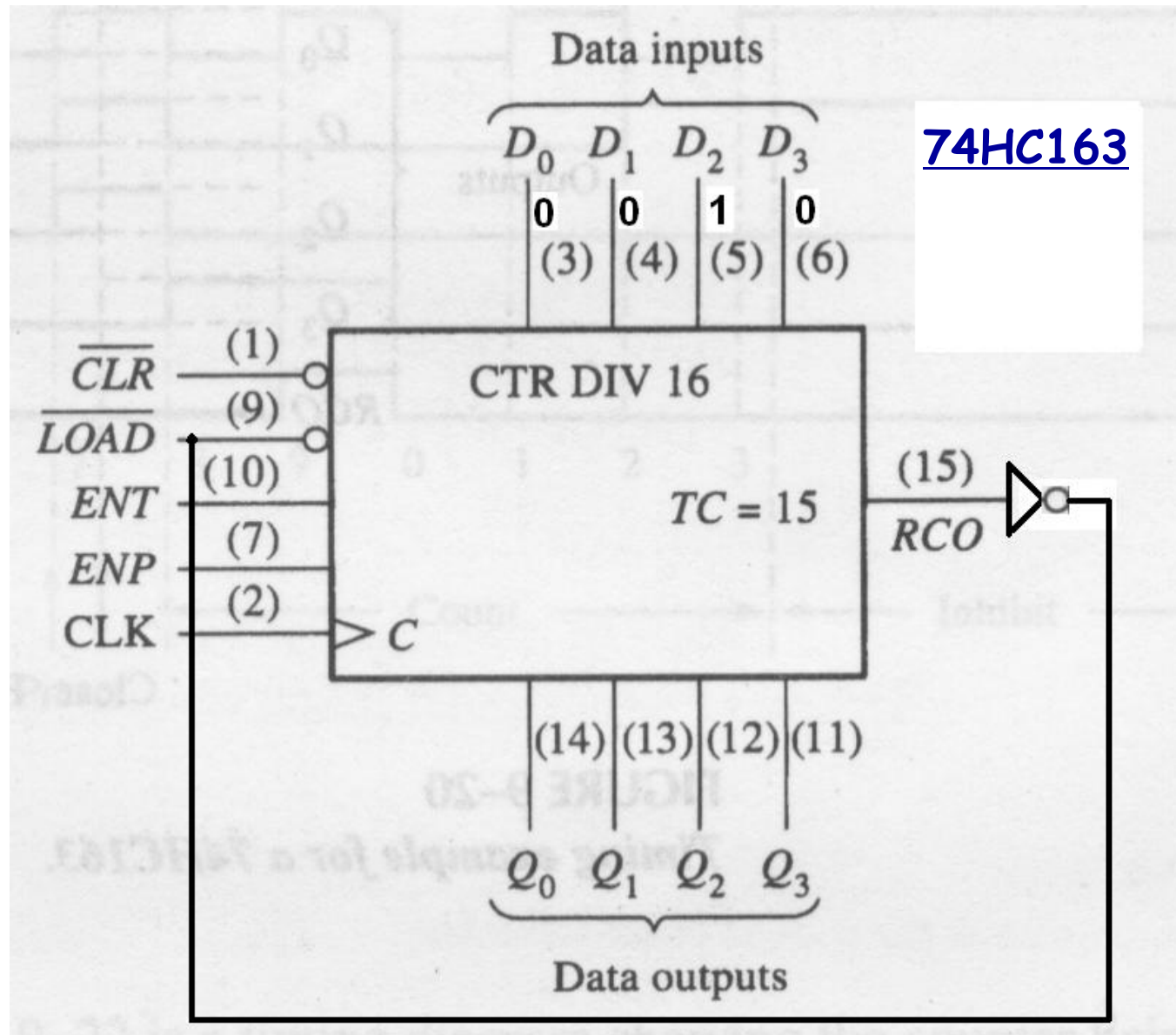


IC Counters with Truncated Sequences

- Basically, only modulus-10 and modulus-16 IC counters are manufactured. We can convert modulus-N IC counter into modulus-M counter by additional logic, where M can be any number less than N.
- For example, a certain application requires a modulus-12 counter. The difference between 16 and 12 is 4, which is the number of states that must be deleted from the full-modulus sequence. We use a technique that is to preset the modulus-16 counter to 4 each time it recycles, so that it will counter from 4 to 15 on each cycle. So each full cycle of the counter consists of 12 states.

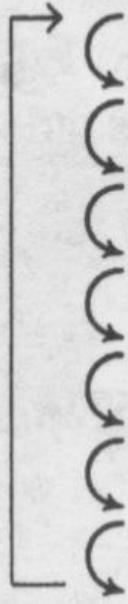
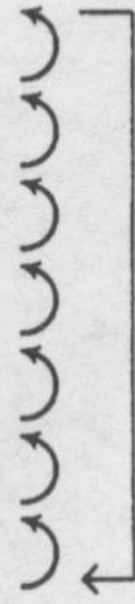


IC Counters with Truncated Sequences



9-4 UP/DOWN SYNCHRONOUS COUNTERS

- An up/down counter is one that is capable of progressing in either direction through a certain sequence.
- An up/down counter, sometimes called a bidirectional counter, can have any specified sequence of states.

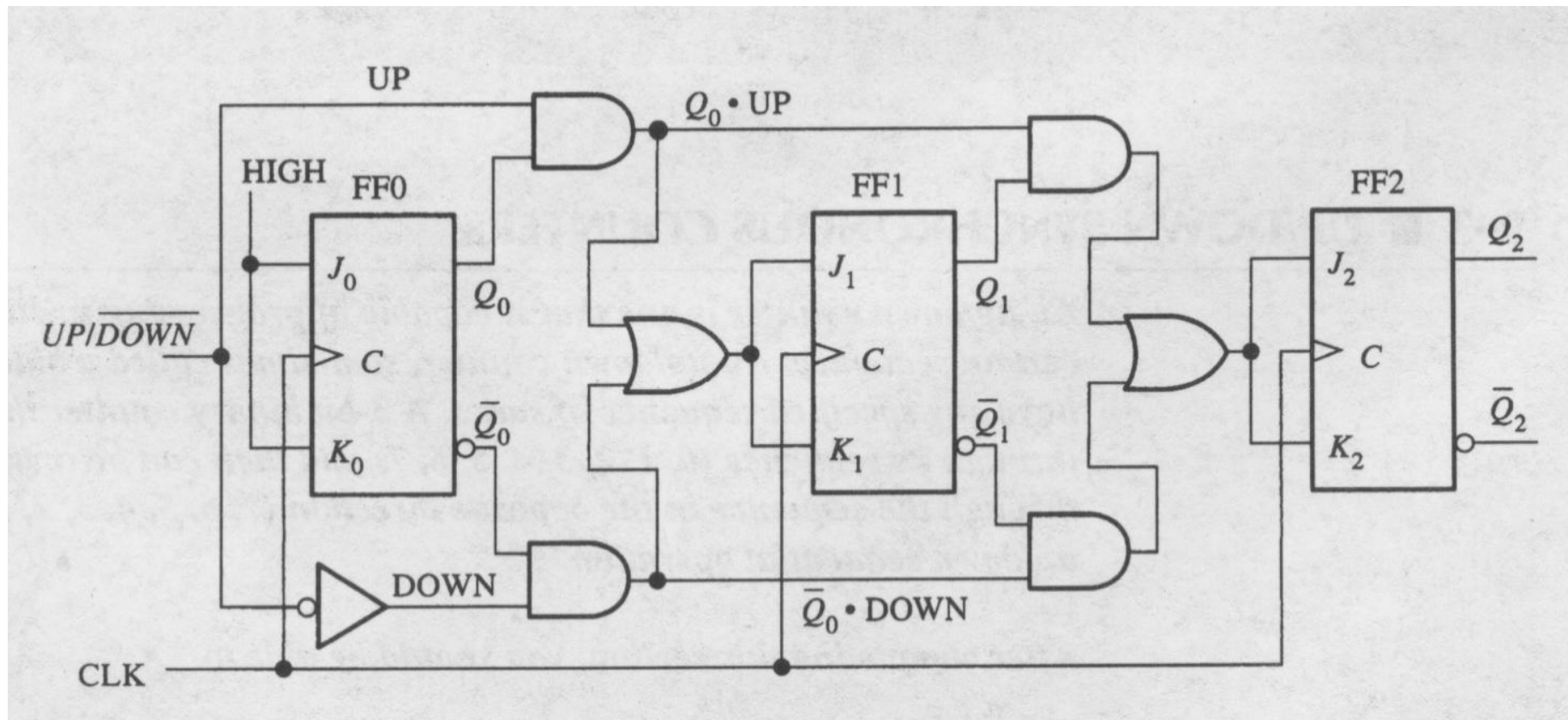
Clock Pulse	UP	$\bar{Q}_2$	$\bar{Q}_1$	$\bar{Q}_0$	DOWN
0		1	1	1	
1		1	1	0	
2		1	0	1	
3		1	0	0	
4		0	1	1	
5		0	1	0	
6		0	0	1	
7		0	0	0	

A 3-bit Up/Down Counter

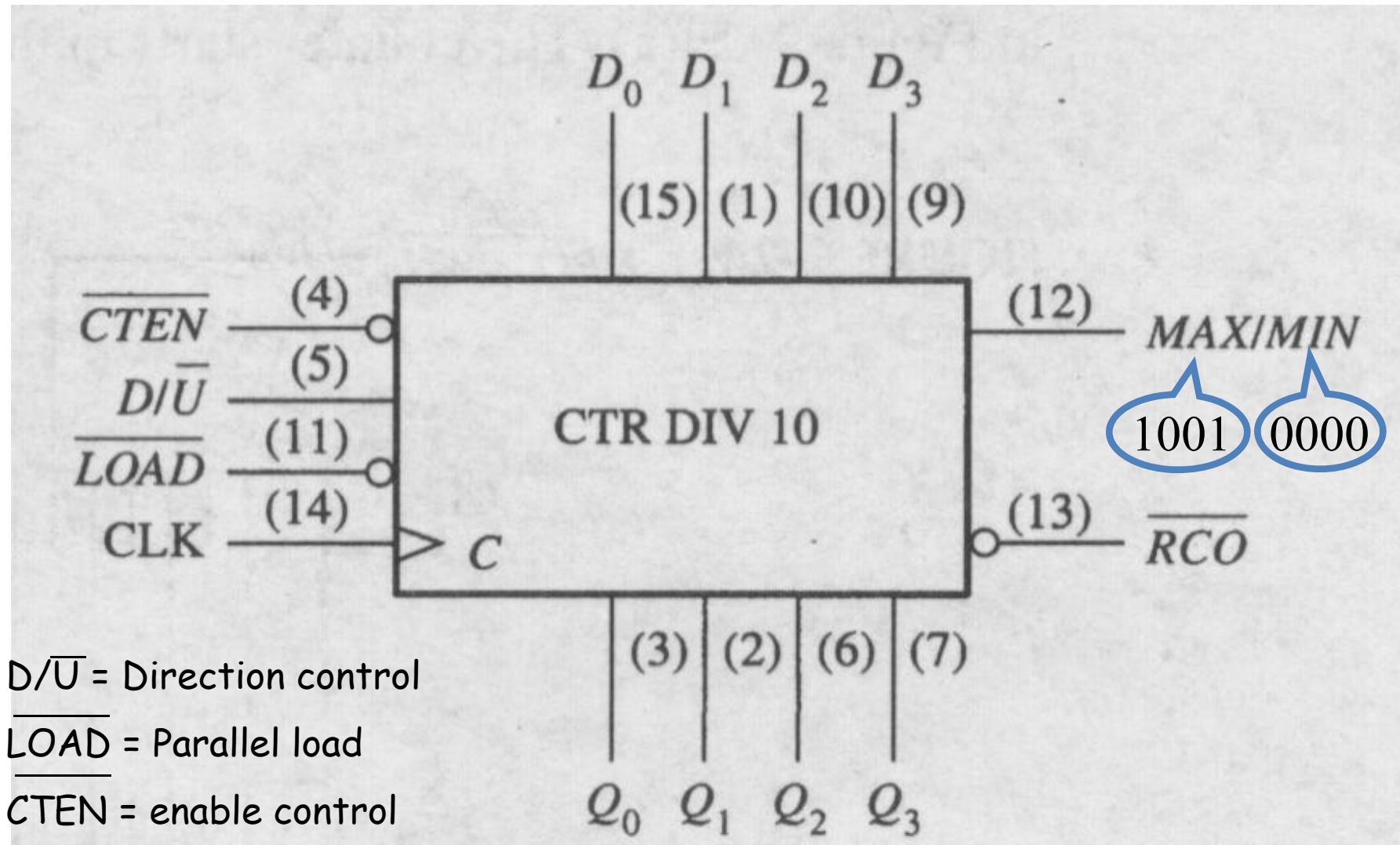
$$J_0 = K_0 = 1$$

$$J_1 = K_1 = Q_0 \cdot UP + \bar{Q}_0 \cdot DOWN$$

$$J_2 = K_2 = Q_0 \cdot Q_1 \cdot UP + \bar{Q}_0 \cdot \bar{Q}_1 \cdot DOWN$$



74LS190: Synchronous Up/Down Decade Counter



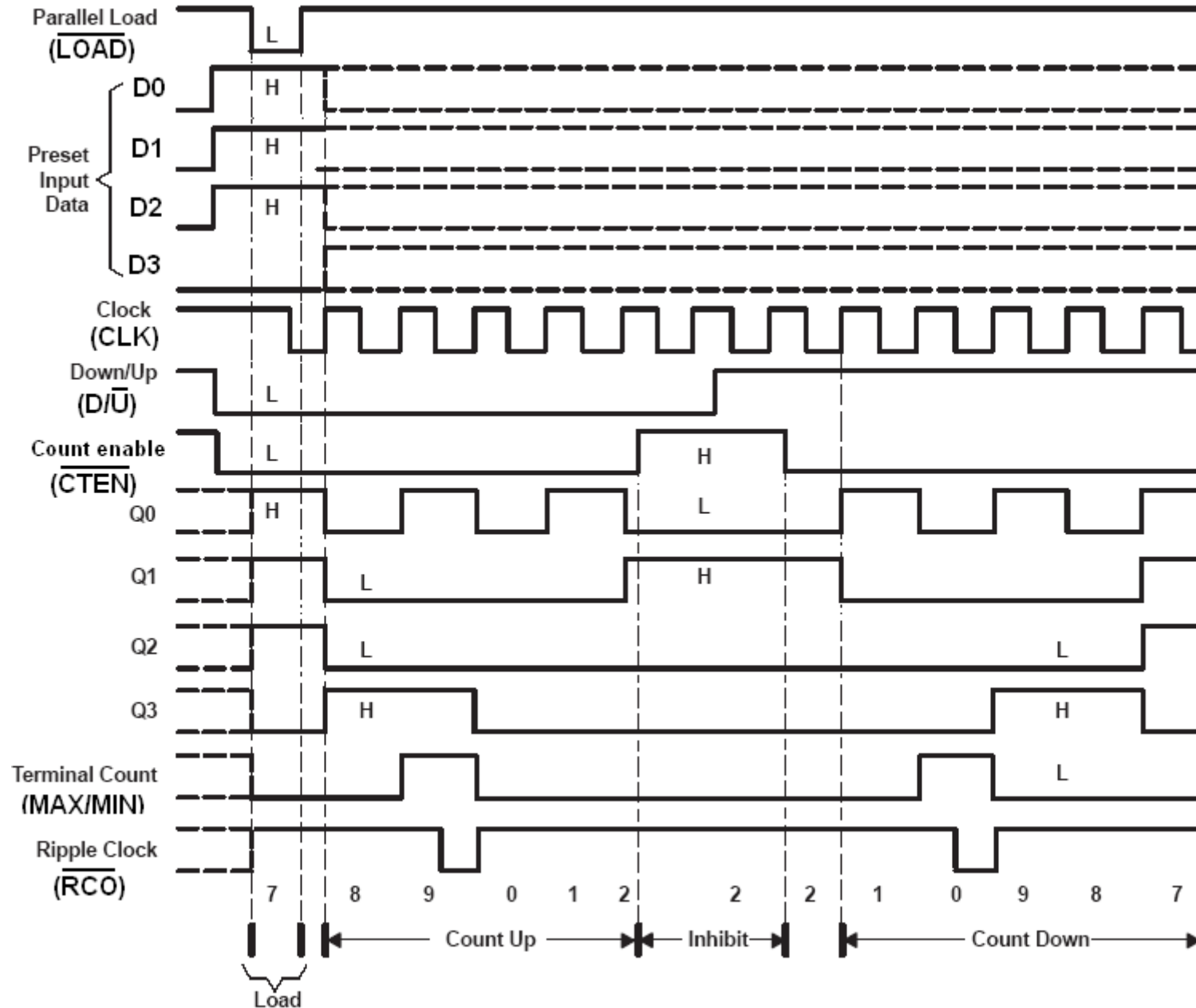
$D/\overline{U}$ = Direction control

$\overline{LOAD}$ = Parallel load

$\overline{CTEN}$ = enable control

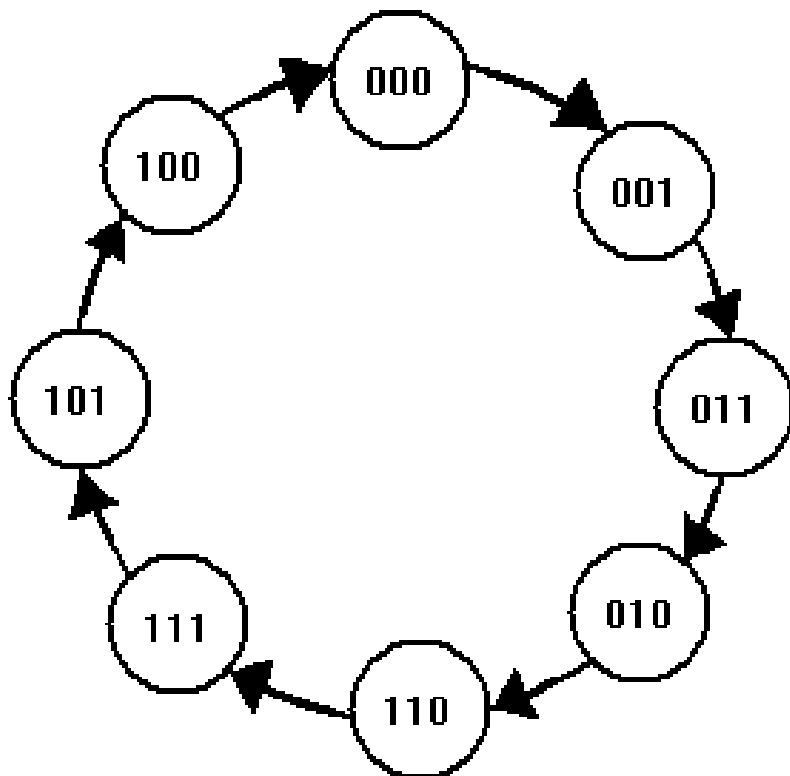
$\overline{RCO}$ = Ripple clock output

Timing Example for 74LS190



9-5 DESIGN OF SYNCHRONOUS COUNTERS

- Counters are a special case of sequential circuits, which shows the progression of states through which the counter advances when it is clocked.
- A counter is first described by a state diagram.
- The second step is to derive a next state table, which lists each state of the counter (present state) along with the corresponding next state.



Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	0	0	0	1
0	0	1	0	1	1
0	1	1	0	1	0
0	1	0	1	1	0
1	1	0	1	1	1
1	1	1	1	0	1
1	0	1	1	0	0
1	0	0	0	0	0

The Characteristic Equation of J-K Flip-Flops

- Assume we use J-K flip-flops. The characteristic equation specifies the flip-flop's next state as a function of its current state and inputs.

Input		Present State		Next State
J	K	Q		Q
0	0	0		0
0	0	1		1
0	1	0		0
0	1	1		0
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		0

Output Transitions		Flip-Flop Inputs	
Q_N	Q_{N+1}	J	K
0	→ 0	0	X
0	→ 1	1	X
1	→ 0	X	1
1	→ 1	X	0

$$Q = J\bar{Q} + \bar{K}Q$$

		KQ			
Q	J	00	01	11	10
	0		1		
	1	1	1		1

The Transition Equations

- From the next state table, we have the following equations which express the next state as function of the current state and the inputs.

$$\begin{aligned}Q_0 &= \overline{Q_2} \overline{Q_1} \overline{Q_0} + \overline{Q_2} \overline{Q_1} Q_0 + Q_2 Q_1 \overline{Q_0} + Q_2 Q_1 Q_0 \\&= (\overline{Q_2} \overline{Q_1} + Q_2 Q_1) \overline{Q_0} + (\overline{Q_2} \overline{Q_1} + Q_2 Q_1) Q_0 \\&= (\overline{Q_2 \oplus Q_1}) \overline{Q_0} + (Q_2 \oplus Q_1) Q_0 \\&= J_0 \overline{Q_0} + \overline{K_0} Q_0\end{aligned}$$

The Transition Equations

- From the next state table, we have the following equations which express the next state as function of the current state and the inputs.

$$\begin{aligned}Q_1 &= \overline{Q}_2 \overline{Q}_1 Q_0 + \overline{Q}_2 Q_1 Q_0 + \overline{Q}_2 Q_1 \overline{Q}_0 + Q_2 Q_1 \overline{Q}_0 \\&= (\overline{Q}_2 Q_0) \overline{Q}_1 + (\overline{Q}_2 Q_0 + \overline{Q}_2 \overline{Q}_0 + Q_2 \overline{Q}_0) Q_1 \\&= (\overline{Q}_2 Q_0) \overline{Q}_1 + (\overline{Q}_2 + \overline{Q}_0) Q_1 \\&= J_1 \overline{Q}_1 + \overline{K}_1 Q_1\end{aligned}$$

The Transition Equations

- From the next state table, we have the following equations which express the next state as function of the current state and the inputs.

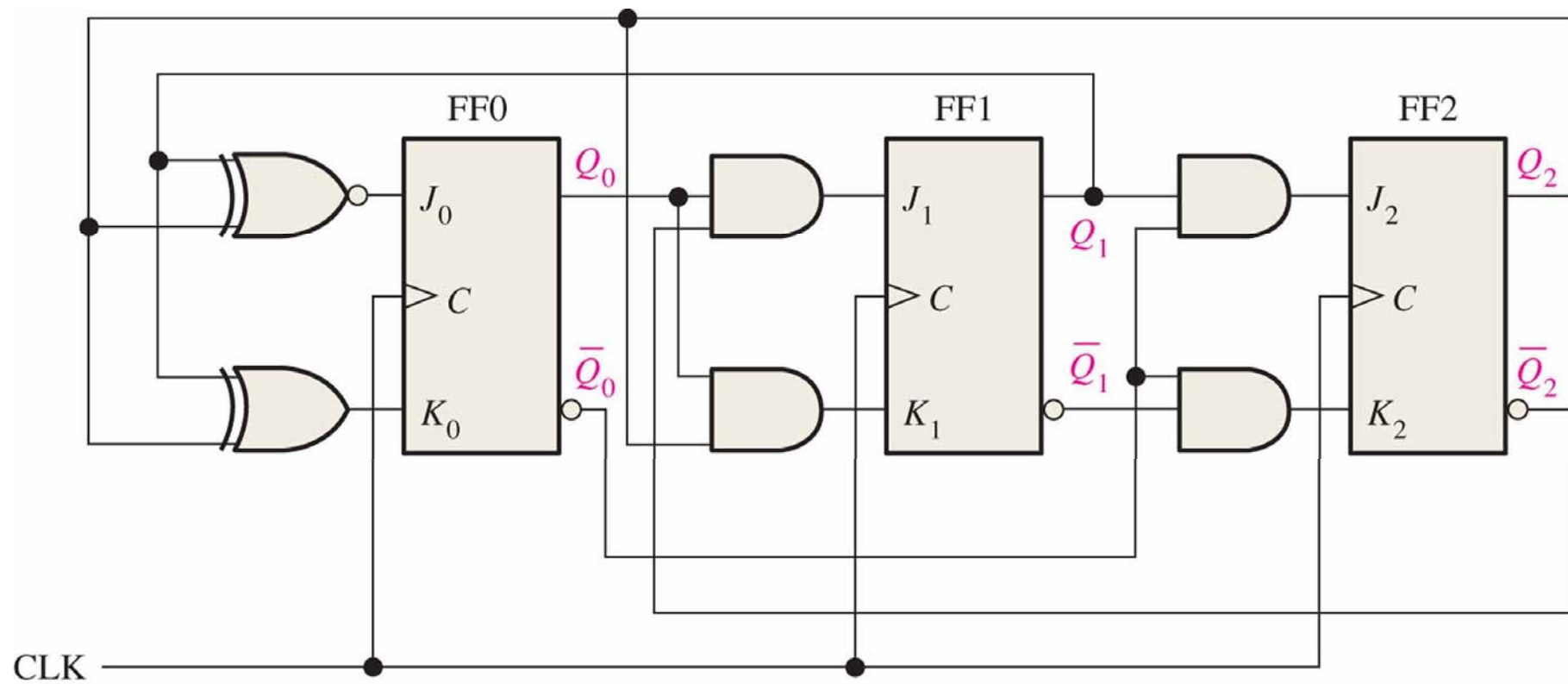
$$\begin{aligned}Q_2 &= \overline{Q_2}Q_1\overline{Q_0} + Q_2Q_1\overline{Q_0} + Q_2Q_1Q_0 + Q_2\overline{Q_1}Q_0 \\&= (Q_1\overline{Q_0})\overline{Q_2} + (Q_1 + Q_0)Q_2 \\&= J_2\overline{Q_2} + \overline{K_2}Q_2\end{aligned}$$

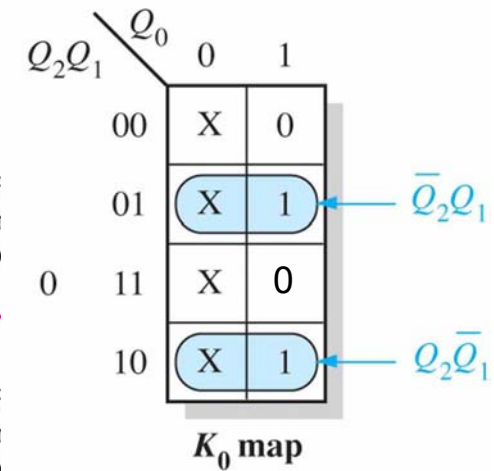
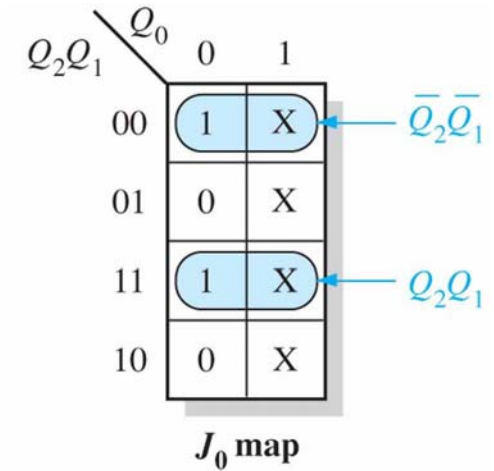
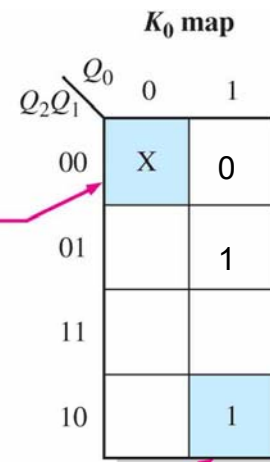
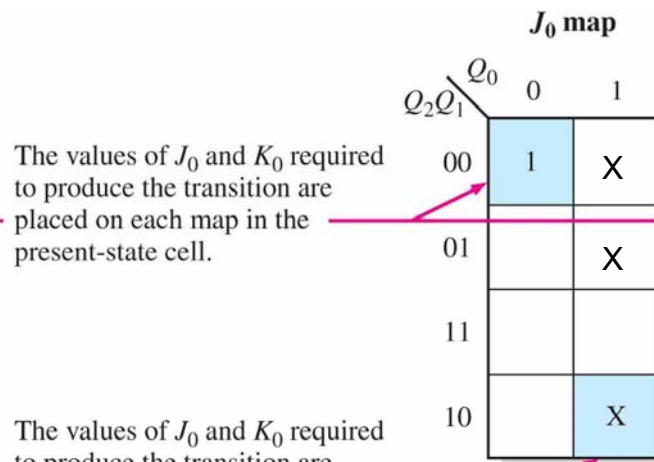
The Excitation Equations

- Logic Equations that express the excitation signals as functions of the current state and input are called excitations and can be derived from the transition equations.

$$\begin{cases} J_2 = Q_1 \bar{Q}_0, K_2 = \overline{Q_1 + Q_0} = \bar{Q}_1 \bar{Q}_0 \\ J_1 = \bar{Q}_2 Q_0, K_1 = \overline{\bar{Q}_2 + \bar{Q}_0} = Q_2 Q_0 \\ J_0 = \overline{Q_2 \oplus Q_1}, K_0 = \overline{\overline{Q_2 \oplus Q_1}} = Q_2 \oplus Q_1 \end{cases}$$

- The final step is to implement the combinational logic from the expressions for the J and K inputs and connect the flip-flops to form the complete 3-bit Gray code counter as shown in Figure 9-31.



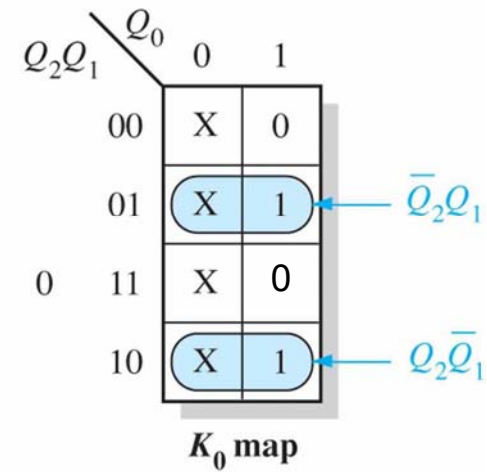
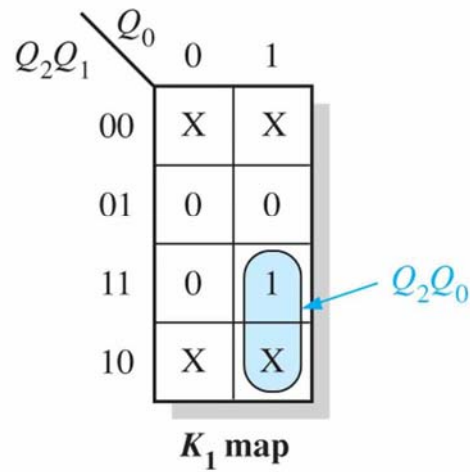
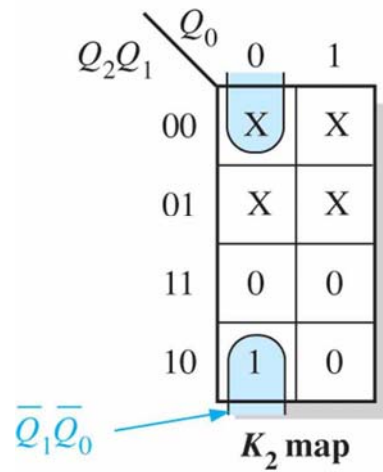
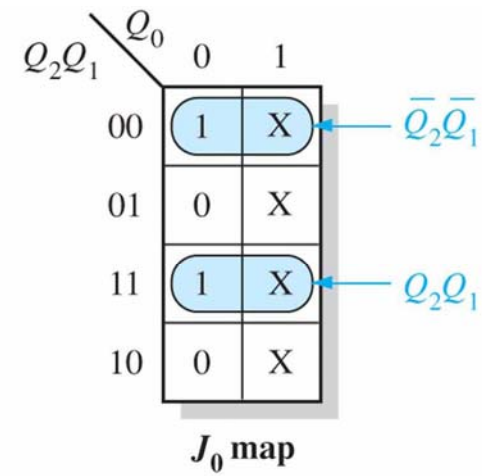
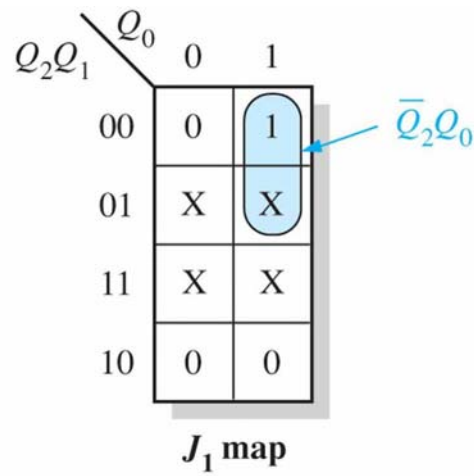
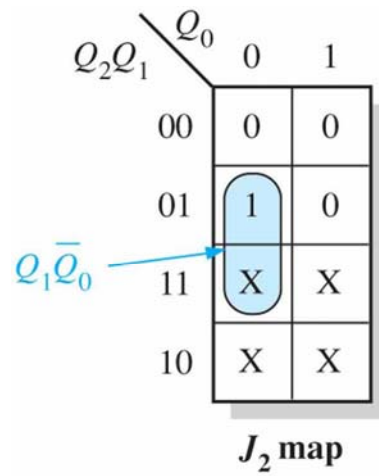


OUTPUT TRANSITIONS		FLIP-FLOP INPUTS	
Q_N	Q_{N+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Flip-flop transition table

PRESENT STATE			NEXT STATE		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	0	0	0	1
0	0	1	0	1	1
0	1	1	0	1	0
0	1	0	1	1	0
1	1	0	1	1	1
1	1	1	1	0	1
1	0	1	1	0	0
1	0	0	0	0	0

Next-state table



$$\begin{cases} J_2 = Q_1 \bar{Q}_0 \\ K_2 = \bar{Q}_1 \bar{Q}_0 \end{cases}$$

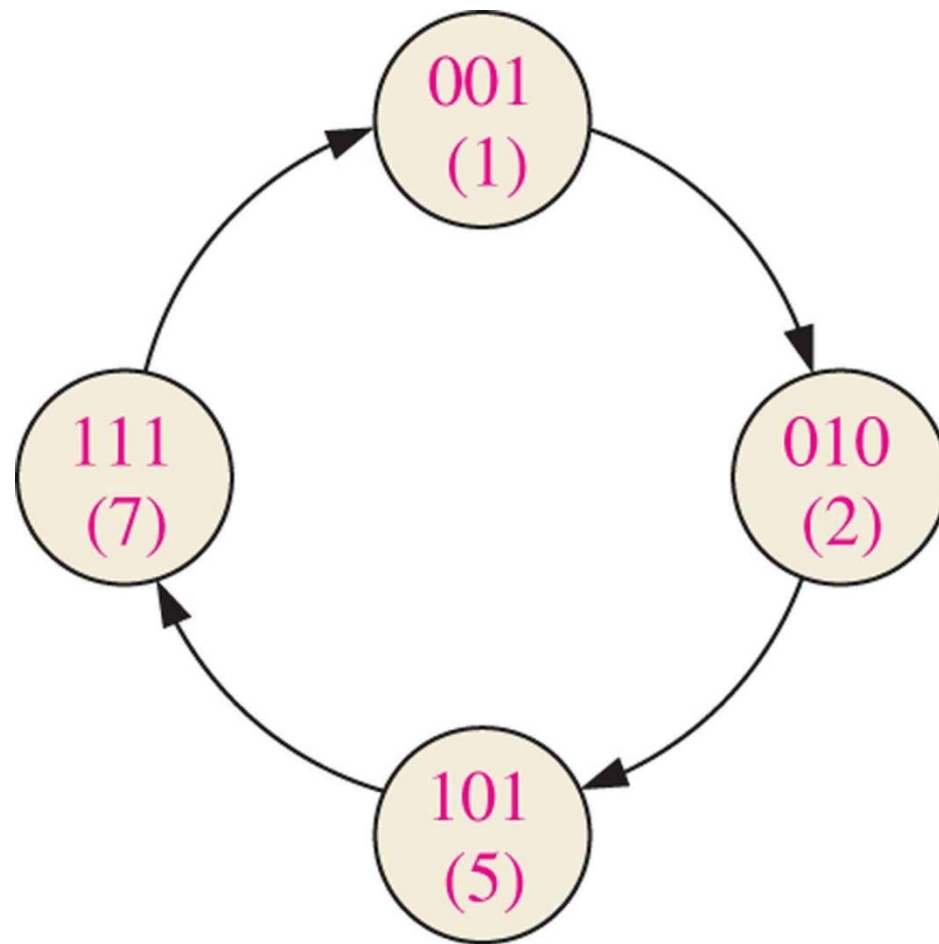
$$\begin{cases} J_1 = \bar{Q}_2 Q_0 \\ K_1 = Q_2 Q_0 \end{cases}$$

$$\begin{cases} J_0 = Q_2 Q_1 + \bar{Q}_2 \bar{Q}_1 = \bar{Q}_2 \oplus \bar{Q}_1 \\ K_0 = \bar{Q}_2 Q_1 + Q_2 \bar{Q}_1 = Q_2 \oplus Q_1 \end{cases}$$

Summarize the procedure of designing counters

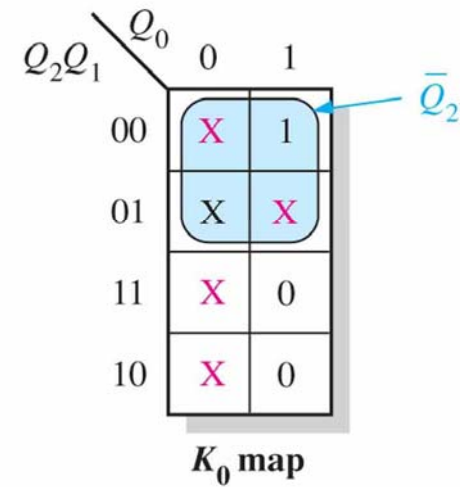
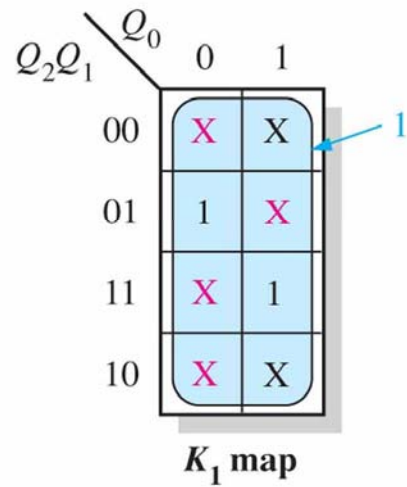
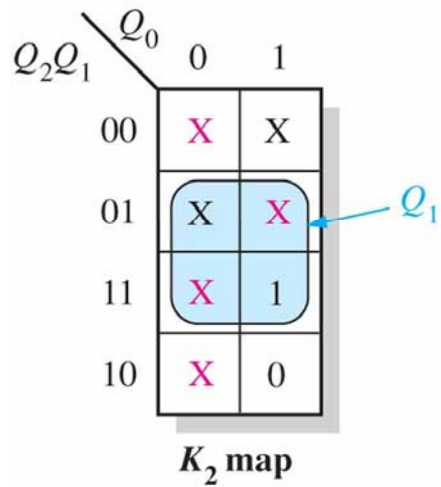
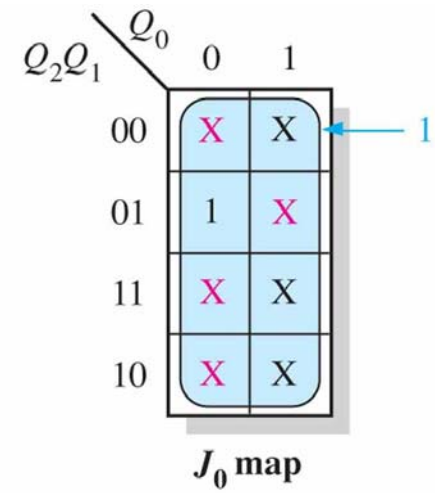
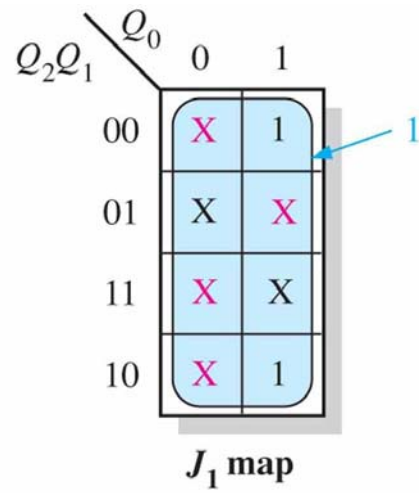
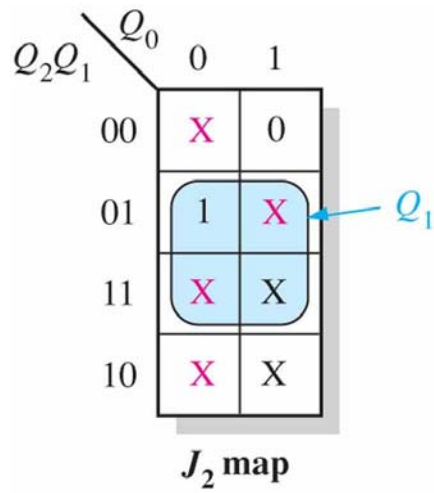
1. Specify the counter sequence and draw a state diagram.
2. Derive a next-step table from the state diagram.
3. Develop a transition table showing the flip-flop inputs required for each transition.
4. Transfer the J and K states from the transition table to Karnaugh maps
5. Group the Karnaugh map cells to generate and derive the logic expression for each flip-flop input.
6. Implement the expressions with combinational logic, and combine with the flip-flops to create the counter.

Example: design a counter with the irregular
binary count sequence



Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	1	0	1	0
0	1	0	1	0	1
1	0	1	1	1	1
1	1	1	0	0	1

Output Transitions Q_N Q_{N+1}		Flip-Flop Inputs J K	
0	→ 0	0	X
0	→ 1	1	X
1	→ 0	X	1
1	→ 1	X	0



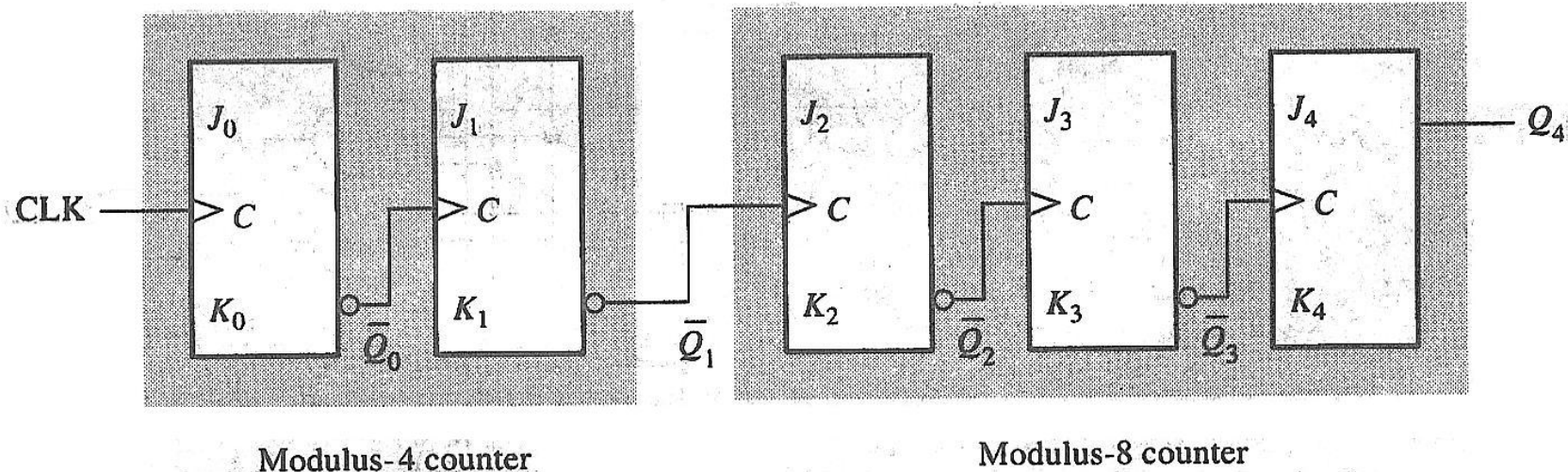
$$\begin{cases} J_2 = Q_1 \\ K_2 = Q_1 \end{cases}$$

$$\begin{cases} J_1 = 1 \\ K_1 = 1 \end{cases}$$

$$\begin{cases} J_0 = 1 \\ K_0 = \bar{Q}_2 \end{cases}$$

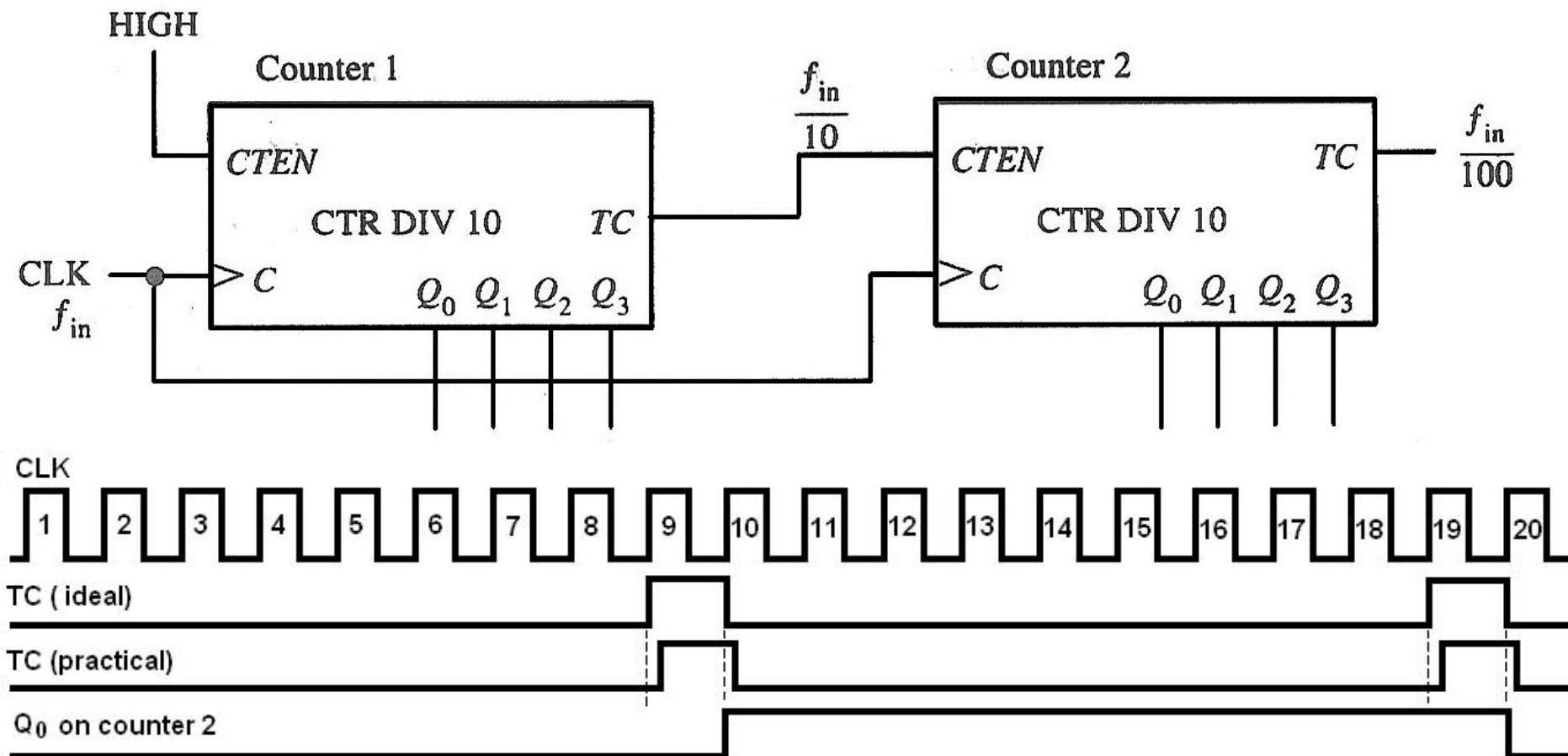
9-6 CASCADED COUNTERS (计数器级联)

- Counters can be connected in cascade to achieve higher-modulus operation. In essence, cascading means that the last-stage output of one counter drives the input of the next counter.
- In general, the overall modulus of cascaded counters is equal to the product of the individual moduli of all the cascaded counters. This can be considered full-modulus



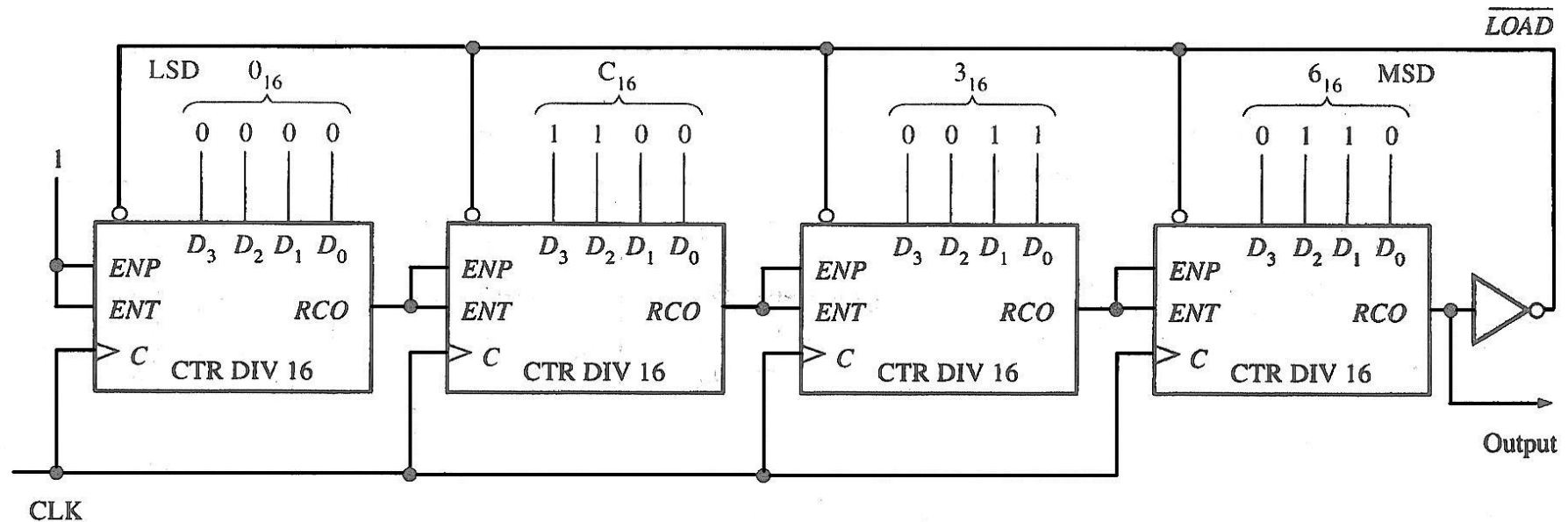
A Modulus-100 Counter Using Two Cascaded Decade Counters

- When operating synchronous counters in a cascaded configuration, it is necessary to use the count enable and the terminal count functions to achieve higher-modulus operation.



Cascaded IC Counters with Truncated Sequences

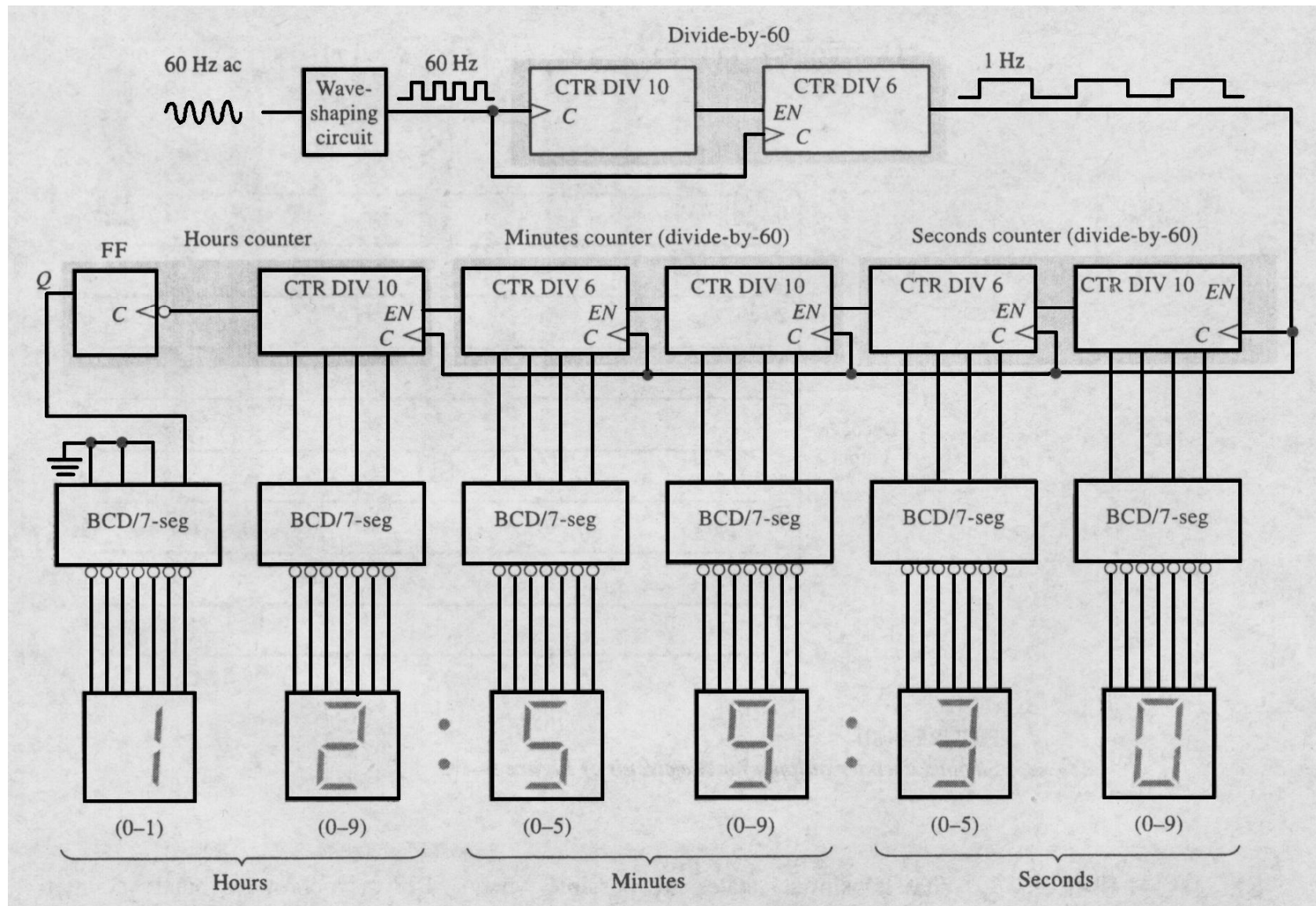
- Often an application requires an overall modulus that is less than that achieved by full-modulus cascading. That is, a truncated sequence must be implemented with cascaded counters.
- Assume that a certain application requires a modulus-40,000 counter. We implement it by modifying the following circuit.



$$\text{MOD } 65,536 \longrightarrow \text{MOD } 40,000: 65,536 - 40,000 = 25,536 = 63C0 \text{ H}$$

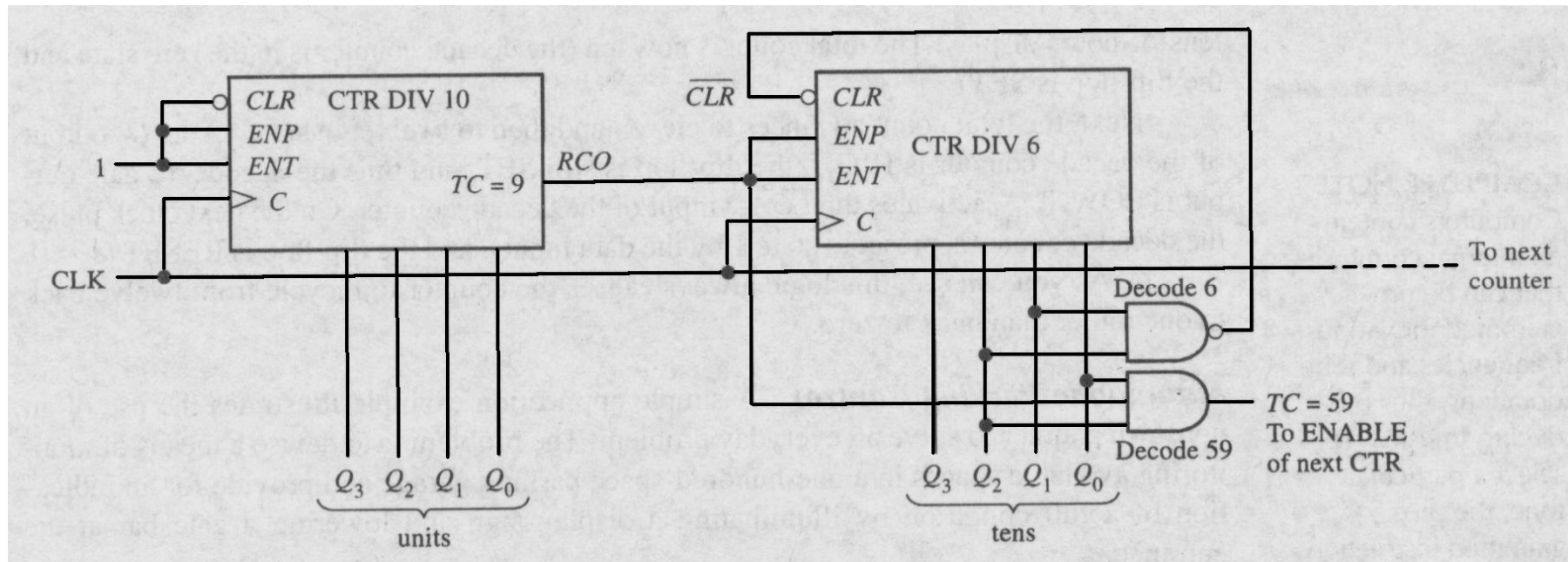
The Digital Clock

- This is a simplified logic diagram of a digital clock that displays seconds, minutes, and hours.



Divide-by-60 Counters

- There are three divide-by-60 counters in the countdown chain.
- Each divide-by-60 counter, implemented by two cascaded 74LS160A synchronous counters, count from 0 to 59 and then recycle to 0.
- The terminal count, 59, is decoded to enable the next counter in the chain.

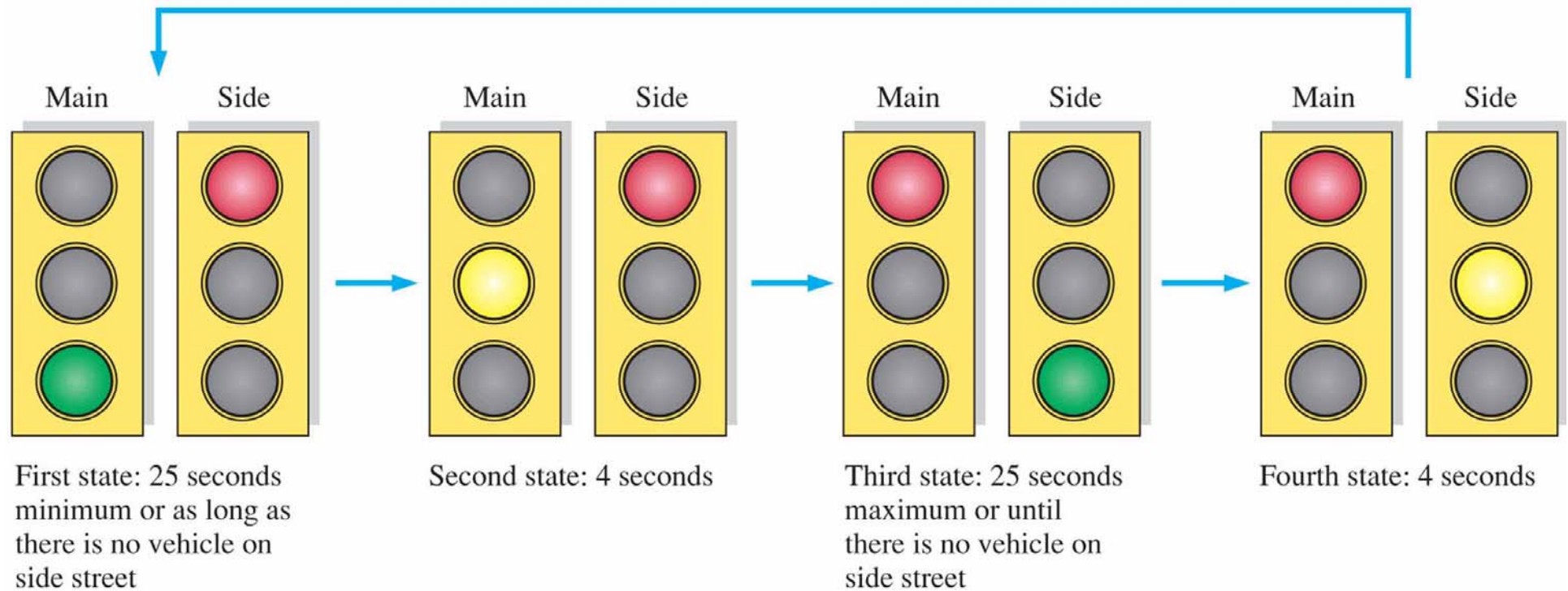


System Application – Traffic Signal

Problem:

Design the **control logic circuits** that controls the sequencing of the lights for *a traffic signal* at the intersection of *a busy main street* and an occasionally used *side street*.

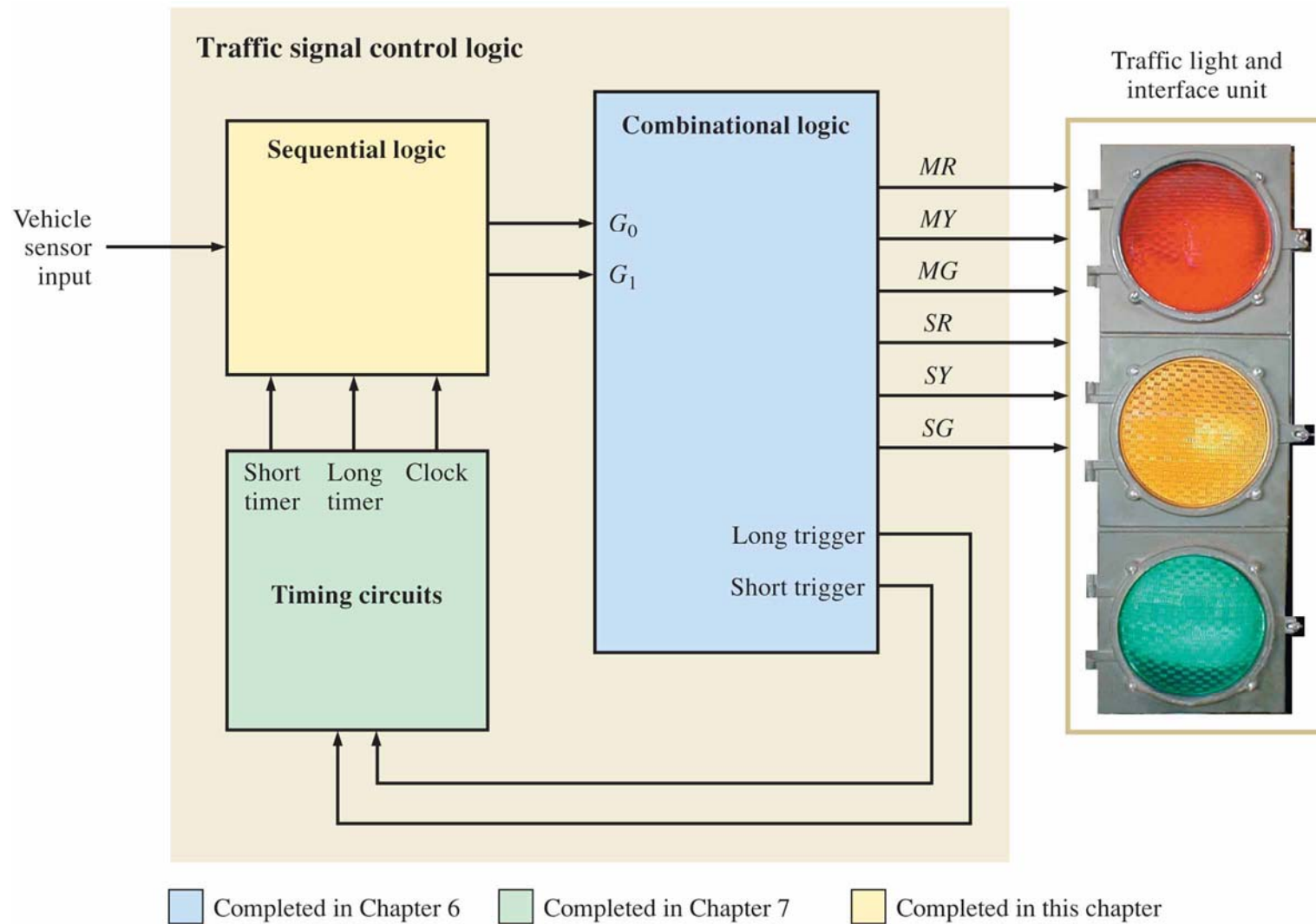
- The green light for the main street will stay on for a minimum of 25s or as long as there is no vehicle on the side street.
- The green light for the side street will stay on until there is no vehicle on the side street or up to a maximum of 25s.
- The yellow caution light will stay on for 4s between changes from green to red on both the main street and the side street.

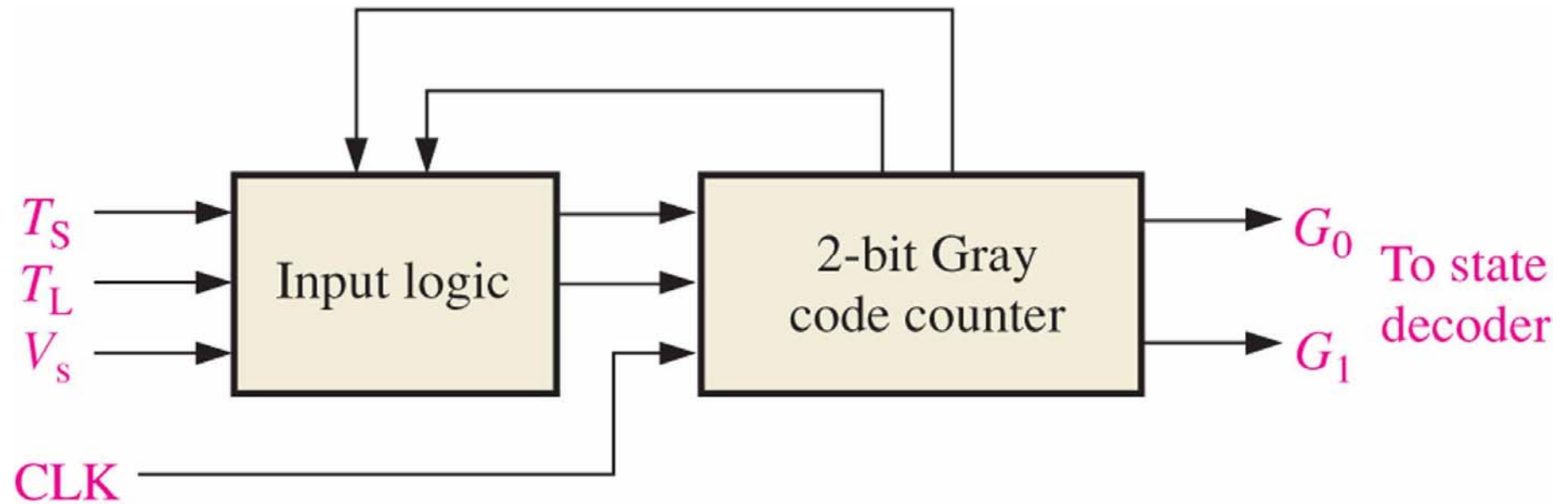


Define some variables:

- V_s : a vehicle is present on the side street.
- T_L : the 25s timer (long timer) is on.
- T_s : the 4s timer (short timer) is on.

Traffic Signal Control: Sequential Part

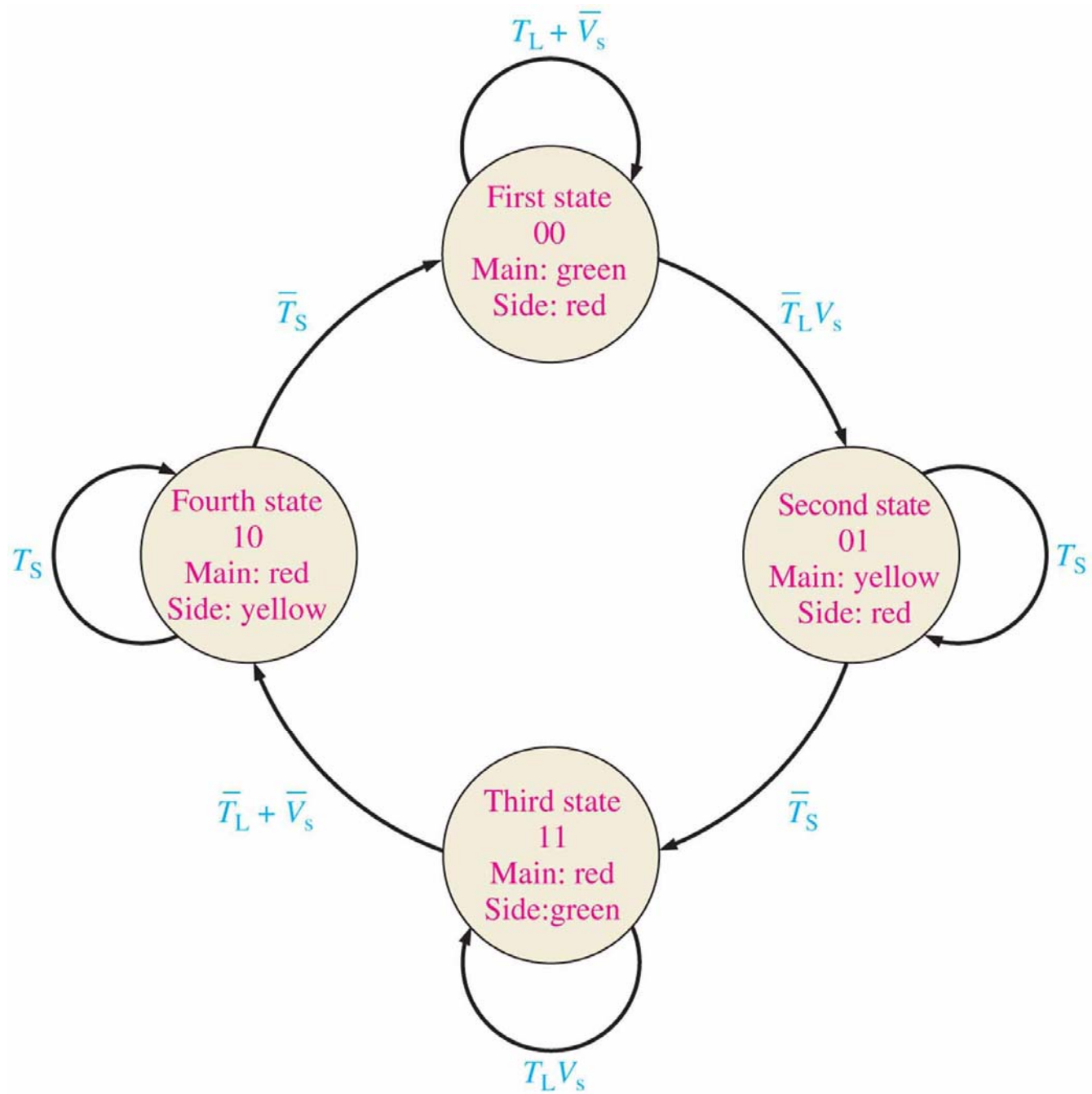


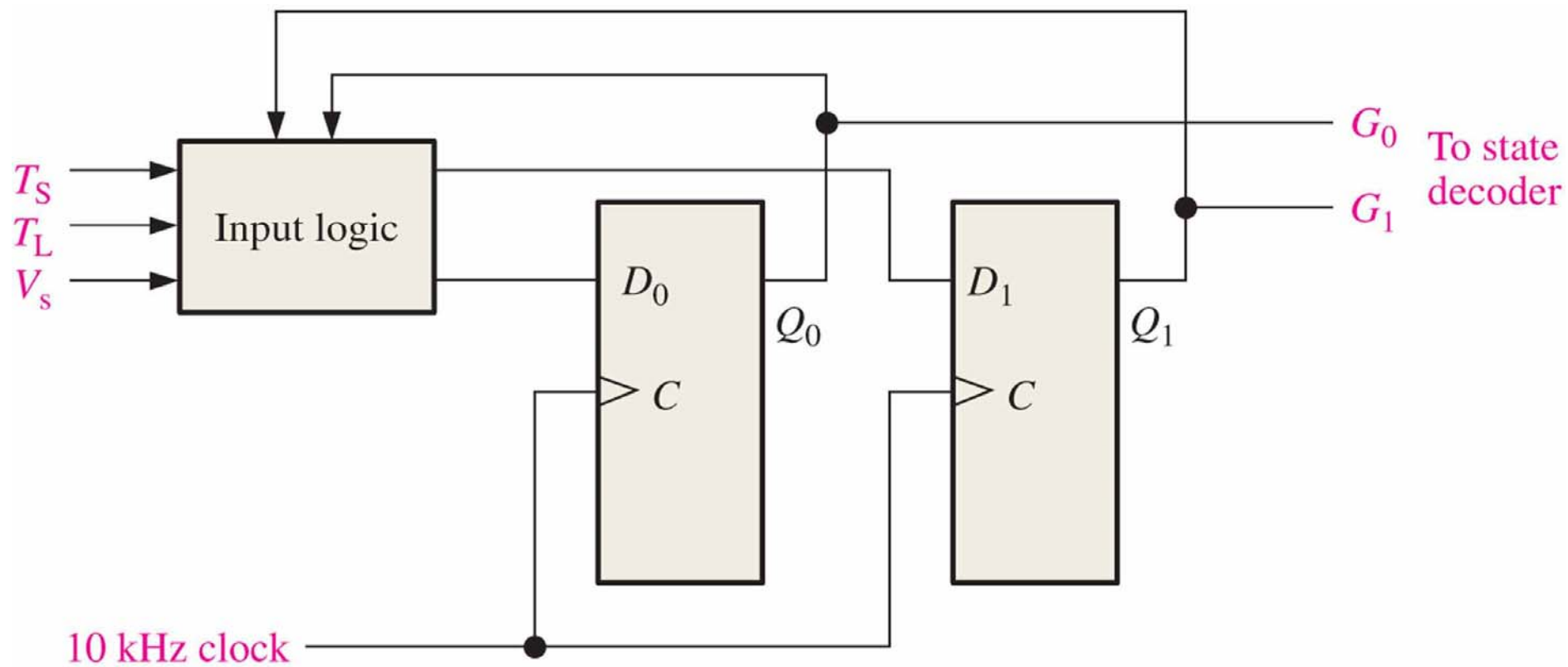


T_S : Short timer (4 s)

T_L : Long timer (25 s)

V_s : Vehicle sensor for the side street



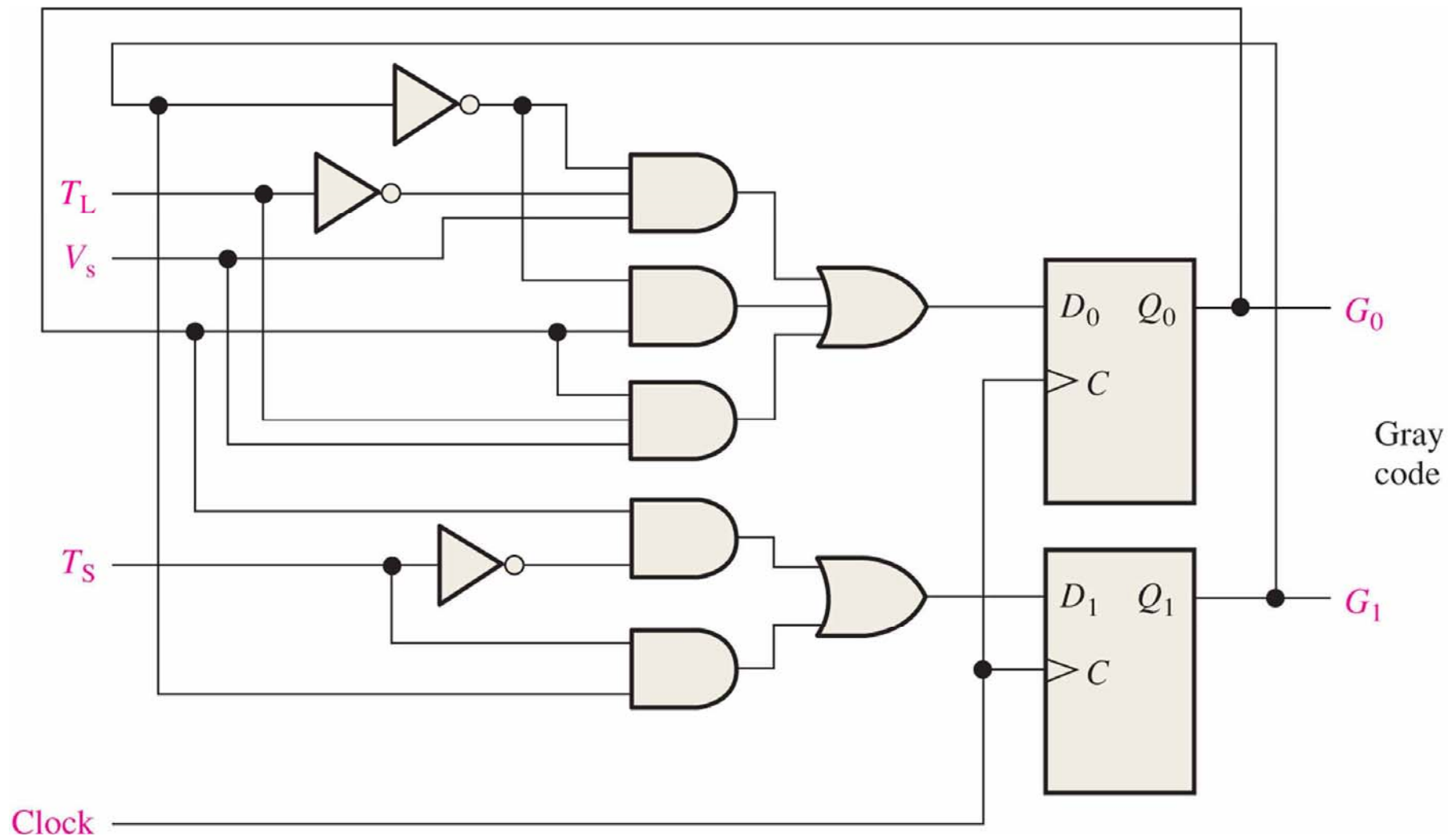


Output Transitions			Flip-Flop Input
Q_N		Q_{N+1}	D
0	$\rightarrow$	0	0
0	$\rightarrow$	1	1
1	$\rightarrow$	0	0
1	$\rightarrow$	1	1

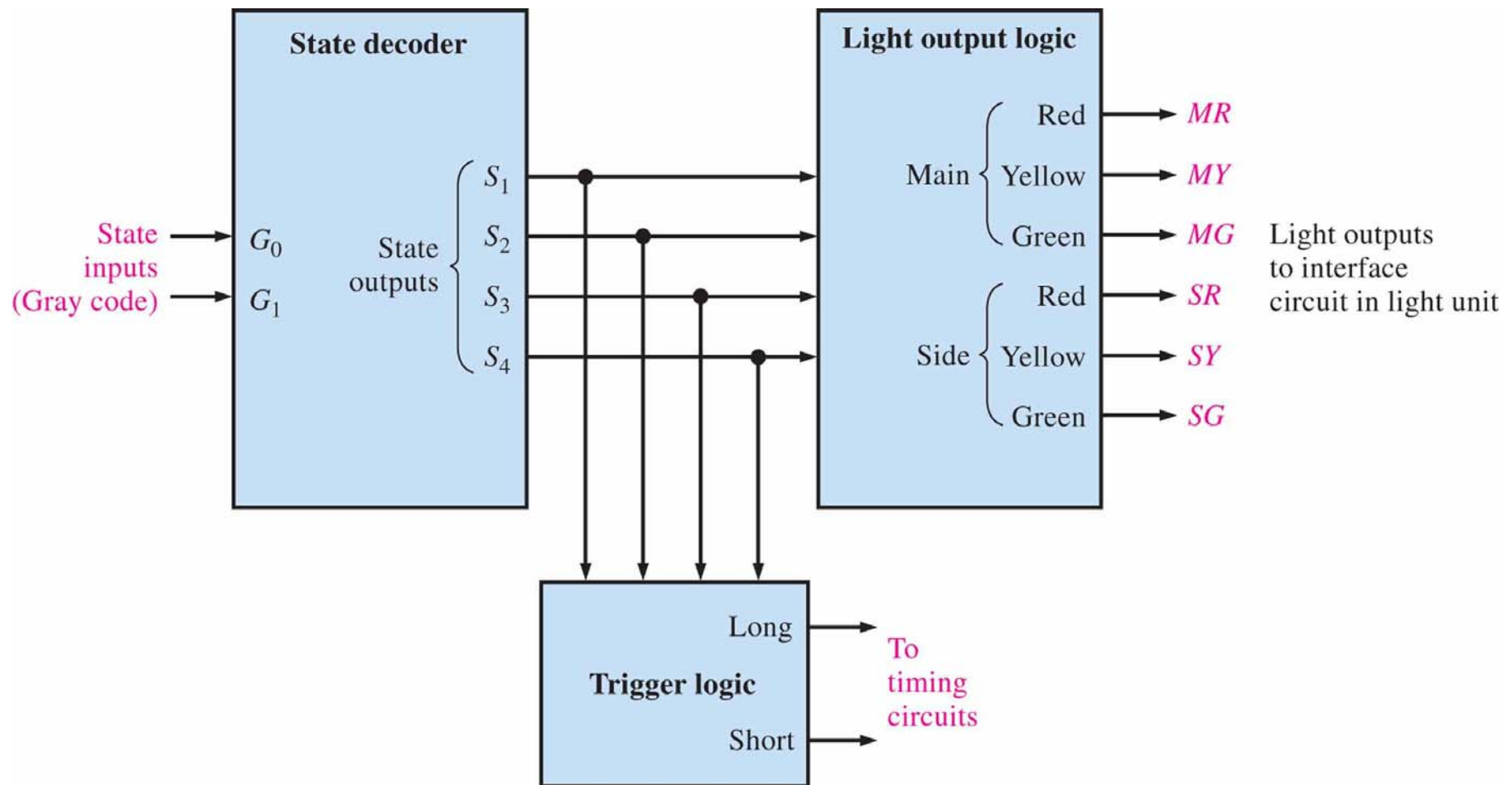
Present State		Next State		Input Conditions	Flip-Flop Input	
Q_1	Q_0	Q_1	Q_0		D	
0	0	0	0	$T_L + \overline{V}_s$	0	0
0	0	0	1	$\overline{T}_L V_s$	0	1
0	1	0	1	T_s	0	1
0	1	1	1	$\overline{T}_s$	1	1
1	1	1	1	$T_L V_s$	1	1
1	1	1	0	$\overline{T}_L + \overline{V}_s$	1	0
1	0	1	0	T_s	1	0
1	0	0	0	$\overline{T}_s$	0	0

$$\begin{aligned}
D_0 &= \overline{Q}_1 \overline{Q}_0 \overline{T}_L V_s + \overline{Q}_1 Q_0 T_S + \overline{Q}_1 Q_0 \overline{T}_S + Q_1 Q_0 T_L V_s \\
&= \overline{Q}_1 \overline{Q}_0 \overline{T}_L V_s + \overline{Q}_1 Q_0 + Q_1 Q_0 T_L V_s
\end{aligned}$$

$$D_1 = \overline{Q}_1 Q_0 \overline{T}_S + Q_1 Q_0 T_L V_s + Q_1 Q_0 \overline{T}_L + Q_1 Q_0 \overline{V}_s + Q_1 \overline{Q}_0 T_S$$



The combinational logic:



State Decoder:

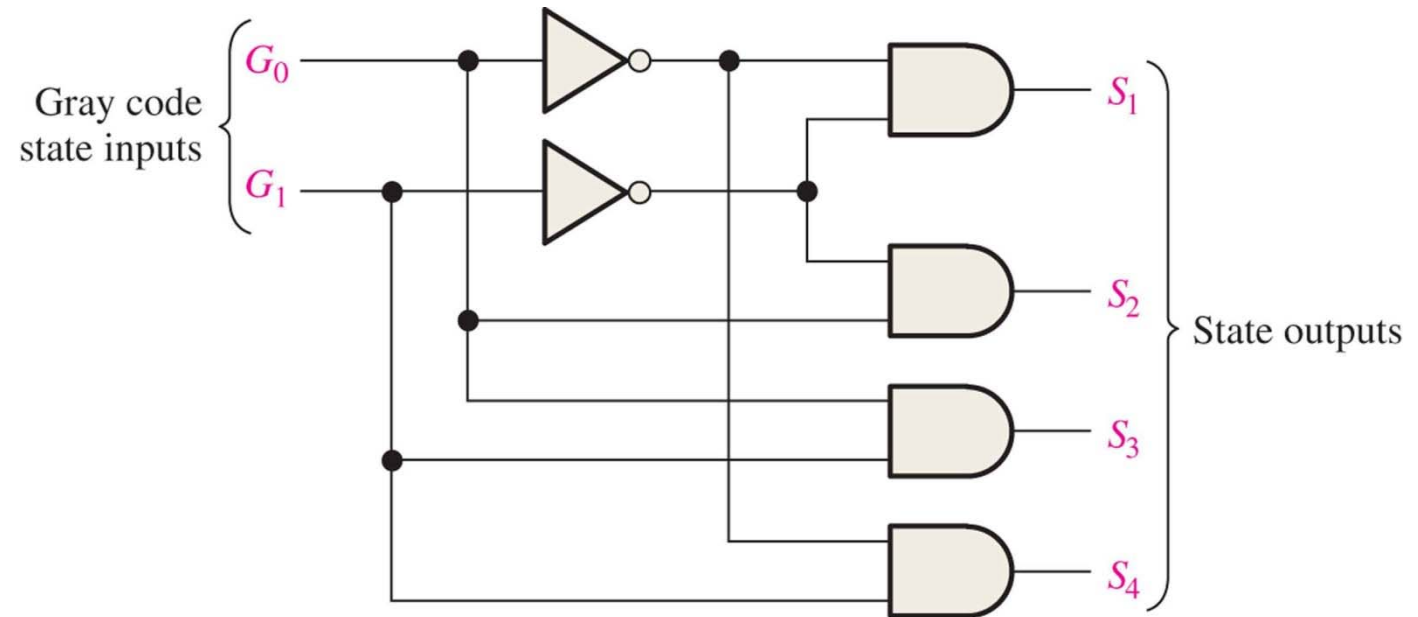
This logic decodes the 2-bit Gray code into four state outputs S_1, S_2, S_3, S_4 .

$$S_1 = \overline{G_1} \overline{G_0}$$

$$S_2 = \overline{G_1} G_0$$

$$S_3 = G_1 G_0$$

$$S_4 = G_1 \overline{G_0}$$



Light Output Logic:

This logic produces six outputs to turn the traffic lights on and off.

MR(main red), MY(main yellow), MG(main green)

SR(side red), SY(side yellow), SG(side green)

$$MR = S_3 + S_4$$

$$MY = S_2$$

$$MG = S_1$$

$$SR = S_1 + S_2$$

$$SY = S_4$$

$$SG = S_3$$

Main Street Lights

Side Street Lights

Trigger Logic:

The trigger logic produces two outputs, the longer trigger output and the short trigger output.

The long trigger output initiates the 25s timer on a LOW-to-HIGH transition.

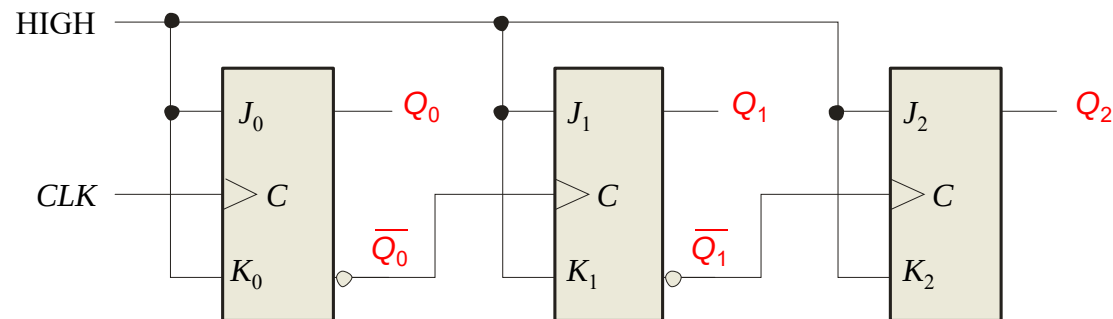
$$LongTrig = S_1 + S_3$$

The short trigger output initiates the 4s timer on a LOW-to-HIGH transition.

$$ShortTrig = S_2 + S_4$$

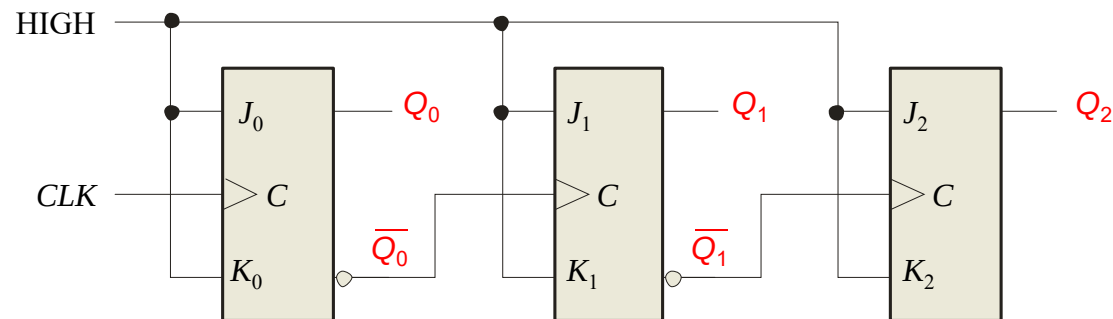
Quiz

1. The counter shown below is an example of
- a. an asynchronous counter
 - b. a BCD counter
 - c. a synchronous counter
 - d. none of the above



Quiz

2. The Q_0 output of the counter shown
- a. is present before Q_1 or Q_2
 - b. changes on every clock pulse
 - c. has a higher frequency than Q_1 or Q_2
 - d. all of the above



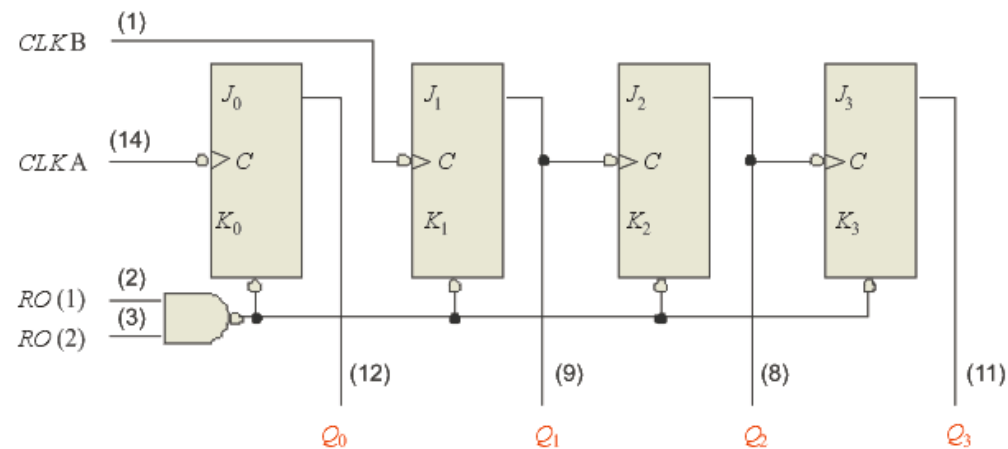
Quiz

3. To cause a D flip-flop to toggle, connect the
- a. clock to the D input
 - b. Q output to the D input
 - c. $\overline{Q}$ output to the D input
 - d. clock to the preset input

Quiz

4. The 7493A asynchronous counter diagram is shown (J 's and K 's are HIGH.) To make the count have a modulus of 16, connect

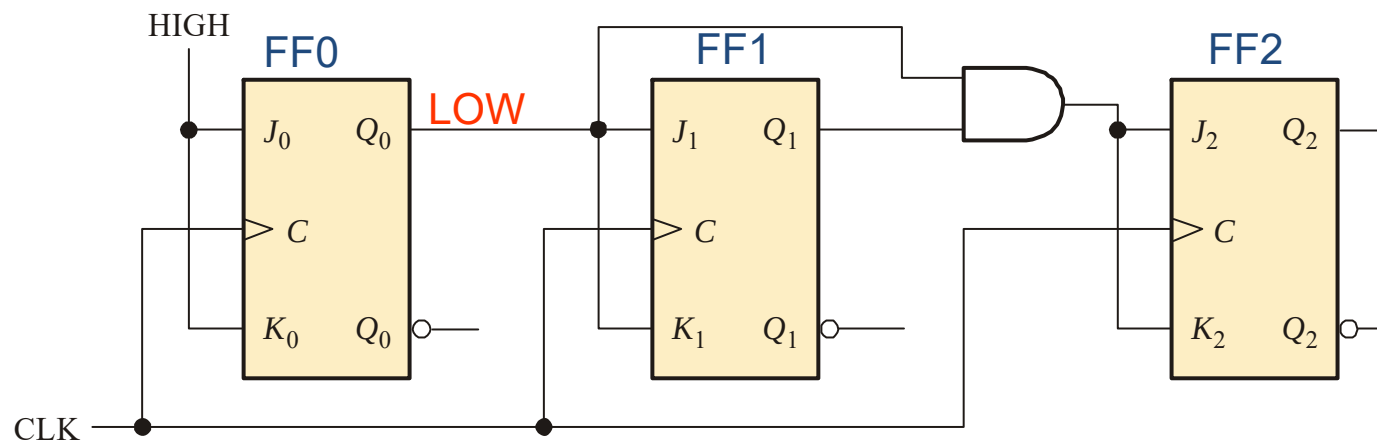
- a. Q_0 to RO(1) and RO(2)
- b. Q_3 to RO(1) and RO(2)
- c. CLK A and CLK B together
- d. Q_0 to CLK B



Quiz

5. Assume Q_0 is LOW. The next clock pulse will cause

- a. FF1 and FF2 to both toggle
- b. FF1 and FF2 to both latch
- c. FF1 to latch; FF2 to toggle
- d. FF1 to toggle; FF2 to latch



Quiz

6. A 4-bit binary counter has a terminal count of

- a. 4
- b. 10
- c. 15
- d. 16

Quiz

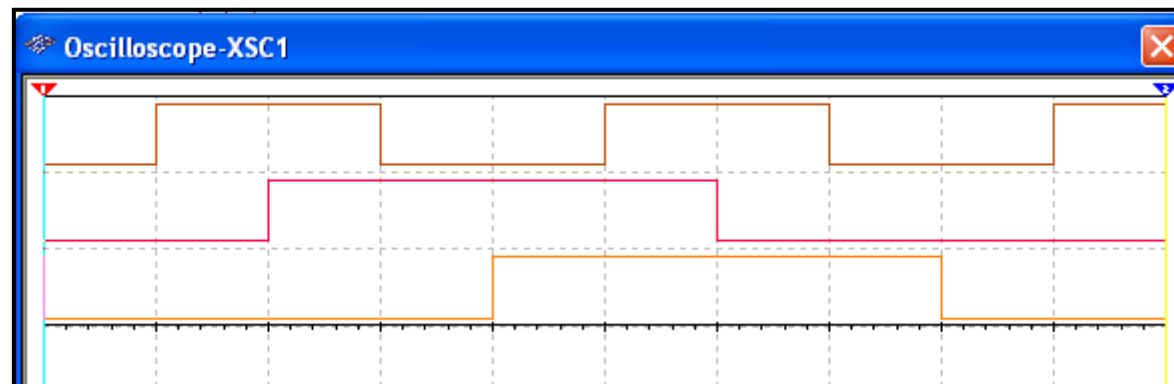
7. Assume the clock for a 4-bit binary counter is 80 kHz. The output frequency of the fourth stage (Q_3) is

- a. 5 kHz
- b. 10 kHz
- c. 20 kHz
- d. 320 kHz

Quiz

8. A 3-bit count sequence is shown for a counter (Q_2 is the MSB). The sequence is

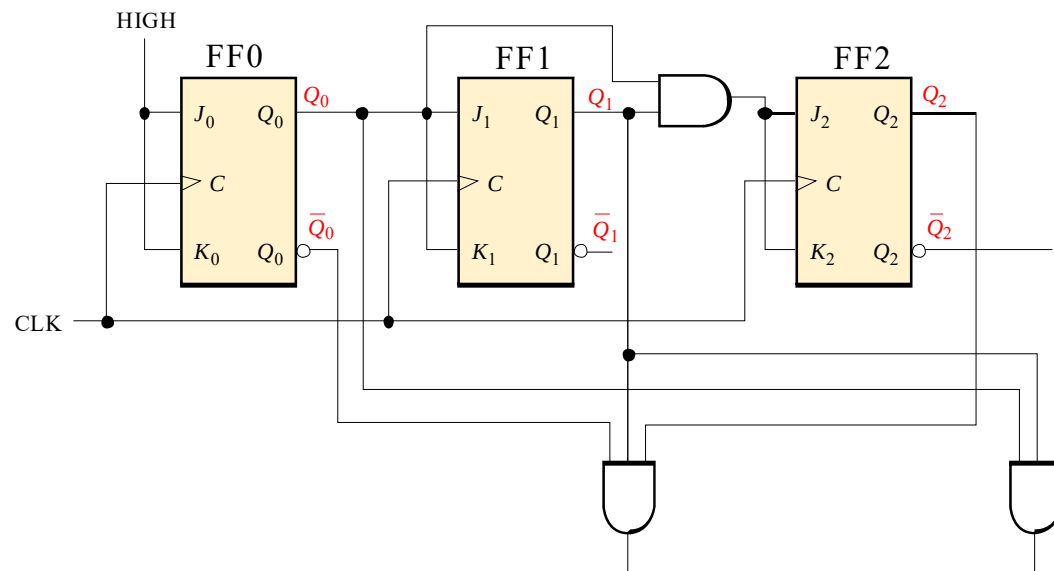
- a. 0-1-2-3-4-5-6-7-0 (repeat)
- b. 0-1-3-2-6-7-5-4-0 (repeat)
- c. 0-2-4-6-1-3-5-7-0 (repeat)
- d. 0-4-6-2-3-7-5-1-0 (repeat)



Quiz

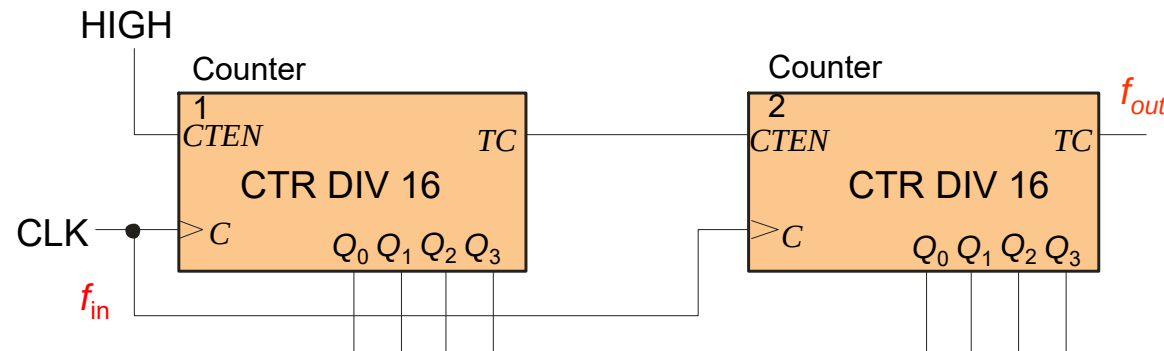
9. FF2 represents the MSB. The counts that are being decoded by the 3-input AND gates are

- 2 and 3
- 3 and 6
- 2 and 5
- 5 and 6



Quiz

- 10. Assume the input frequency (f_{in}) is 256 kHz. The output frequency (f_{out}) will be
 - a. 16 kHz
 - b. 1 kHz
 - c. 65 kHz
 - d. none of the above



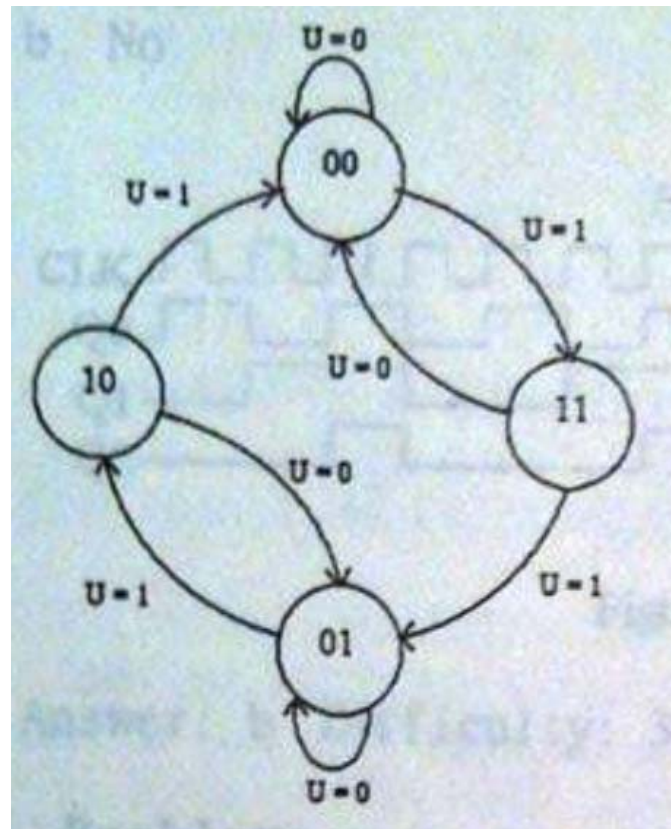
Homework

- Problems: 4, 6, 8, 9, 11, 12, 15, 18, 19, 22, 23, 24(c), 35
- External problem (see next page)



External problem

Design a circuit using J-K flip flops that will perform the count sequence of the state diagram shown below.



Answers

1. a 6. c

2. d 7. a

3. c 8. b

4. d 9. b

5. b 10. b

Hours Counter and Decoders

- Initially both the decade counter and the flip-flop are RESET. The decade counter advances all of its states from 0 to 9, and on the clock pulse that recycles it from 9 back to 0, the flip-flop goes to the SET state. Next, the total counter advances to 11 and then to 12. On the next clock pulse, the decade counter is preset to state 1.

